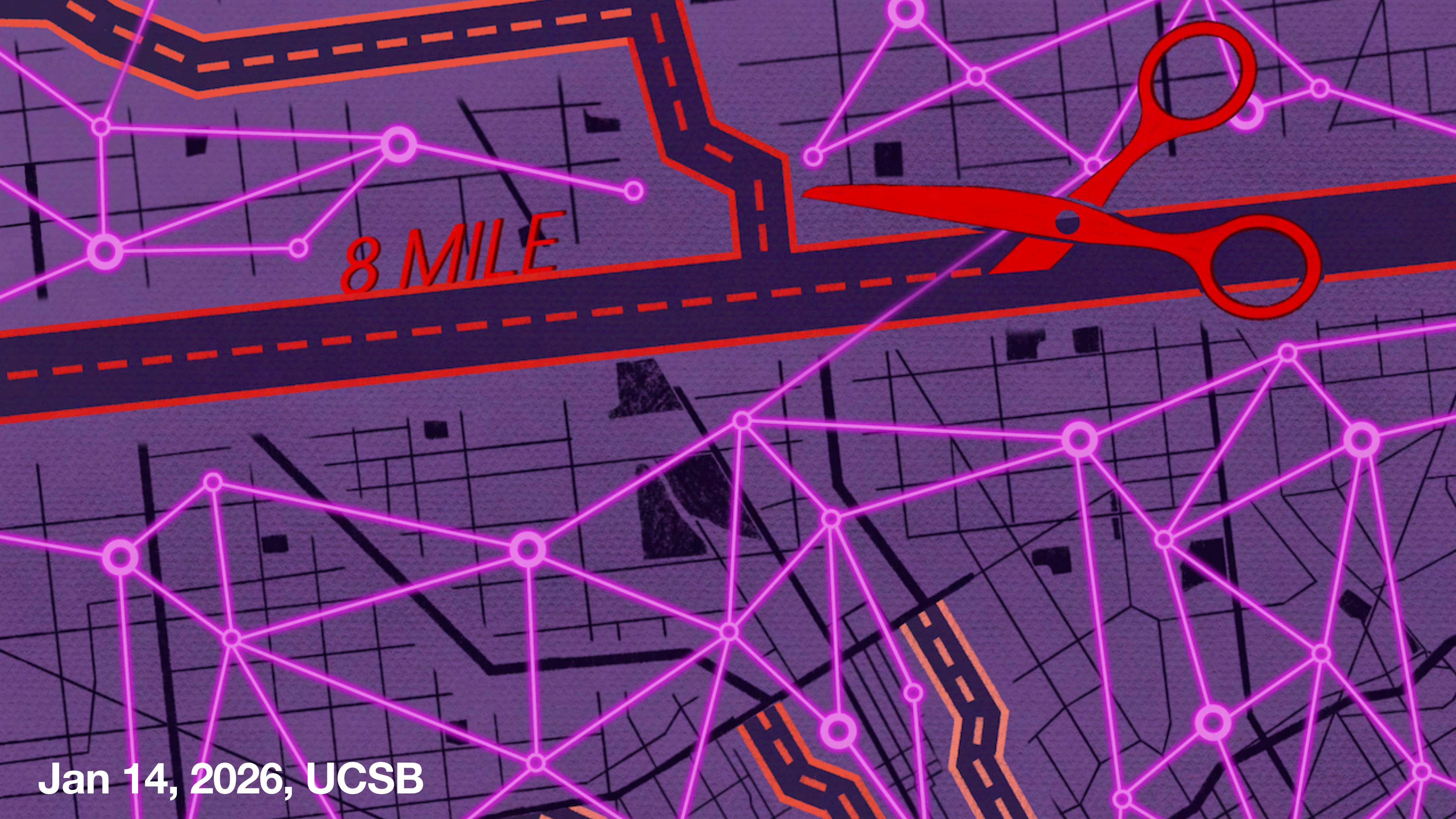
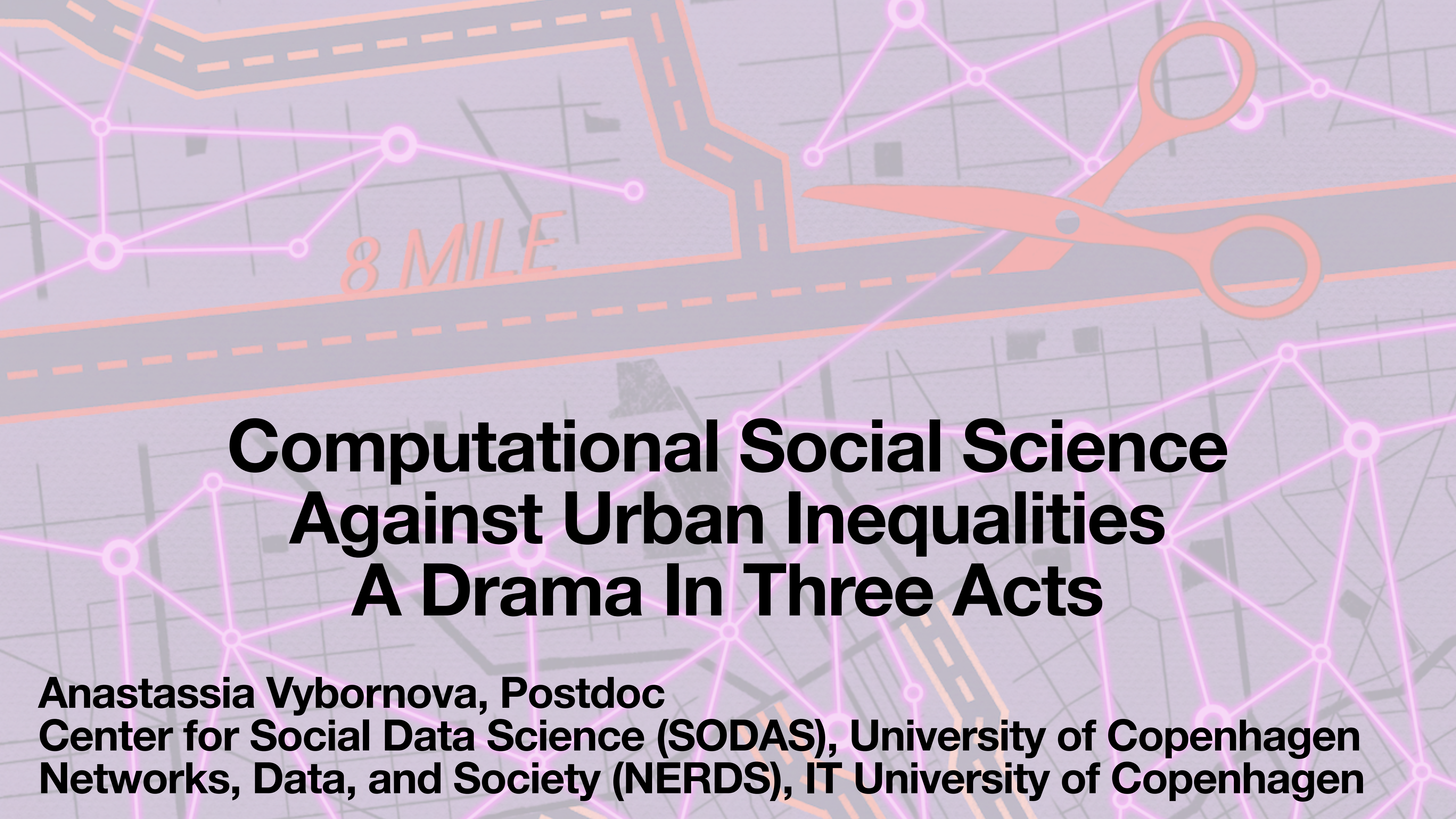




Jan 14, 2026, UCSB



Computational Social Science Against Urban Inequalities A Drama In Three Acts

**Anastassia Vybornova, Postdoc
Center for Social Data Science (SODAS), University of Copenhagen
Networks, Data, and Society (NERDS), IT University of Copenhagen**

Act I: Urban Highways Are Barriers To Social Ties

Act II: You Don't Have To Live Next To Me

Act III: Data for White Supremacy The case of the Danish 'Parallel Societies Index'

Act IV: A Magic Trick

Act I: Urban Highways Are Barriers To Social Ties

Urban highways are barriers to social ties

Luca Maria Aiello^{a,b,1} , Anastassia Vybornova^a , Sándor Juhász^{c,d,e} , Michael Szell^{a,b,c,f} , and Eszter Bokányi^g

Affiliations are included on p. 9.

Edited by Susan Hanson, Clark University, Worcester, MA; received May 7, 2024; accepted January 23, 2025

IT UNIVERSITY OF COPENHAGEN



UNIVERSITEIT
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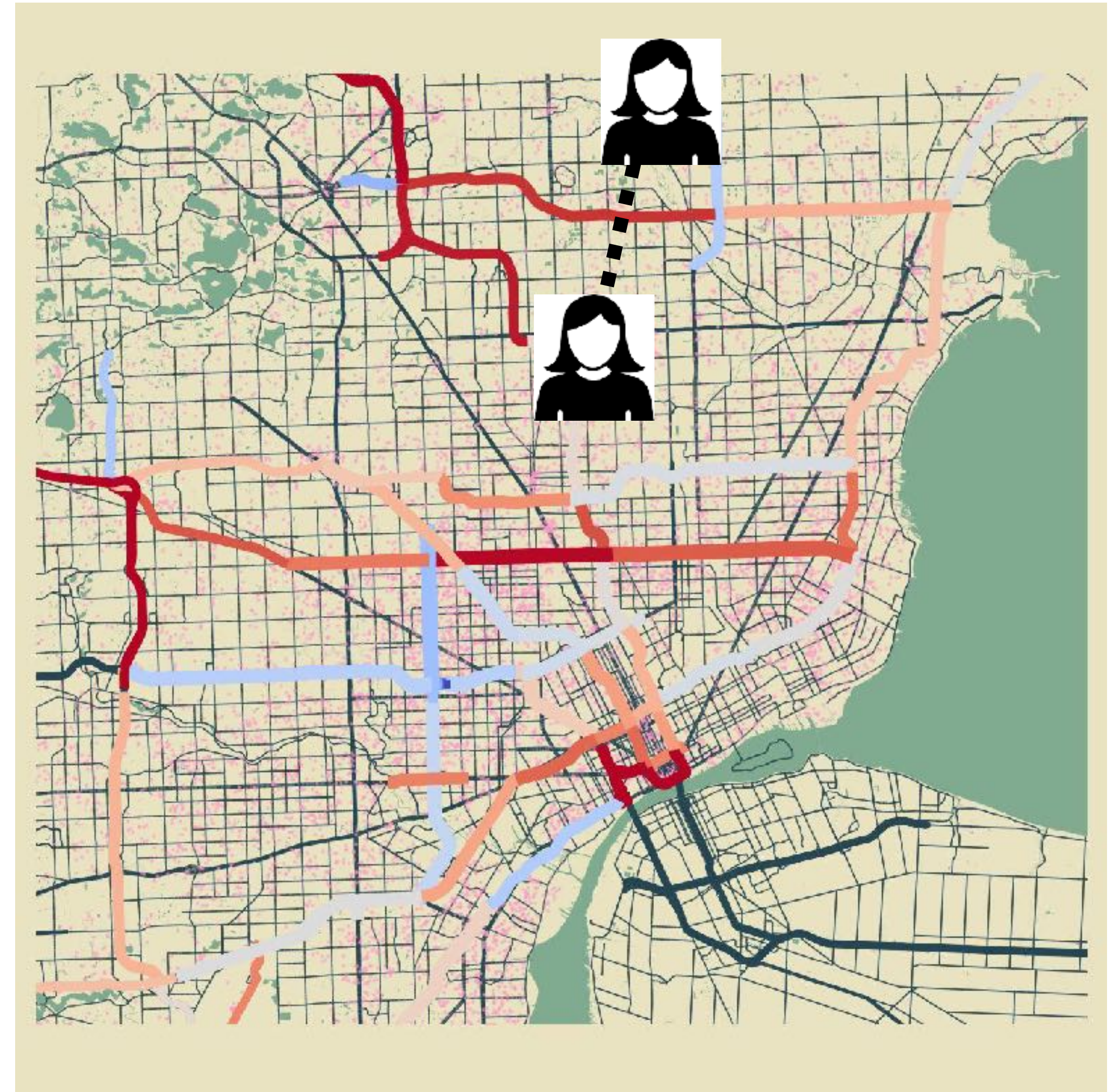


BUDAPESTI
CORVINUS
EGYETEM

<https://www.pnas.org/doi/10.1073/pnas.2408937122>

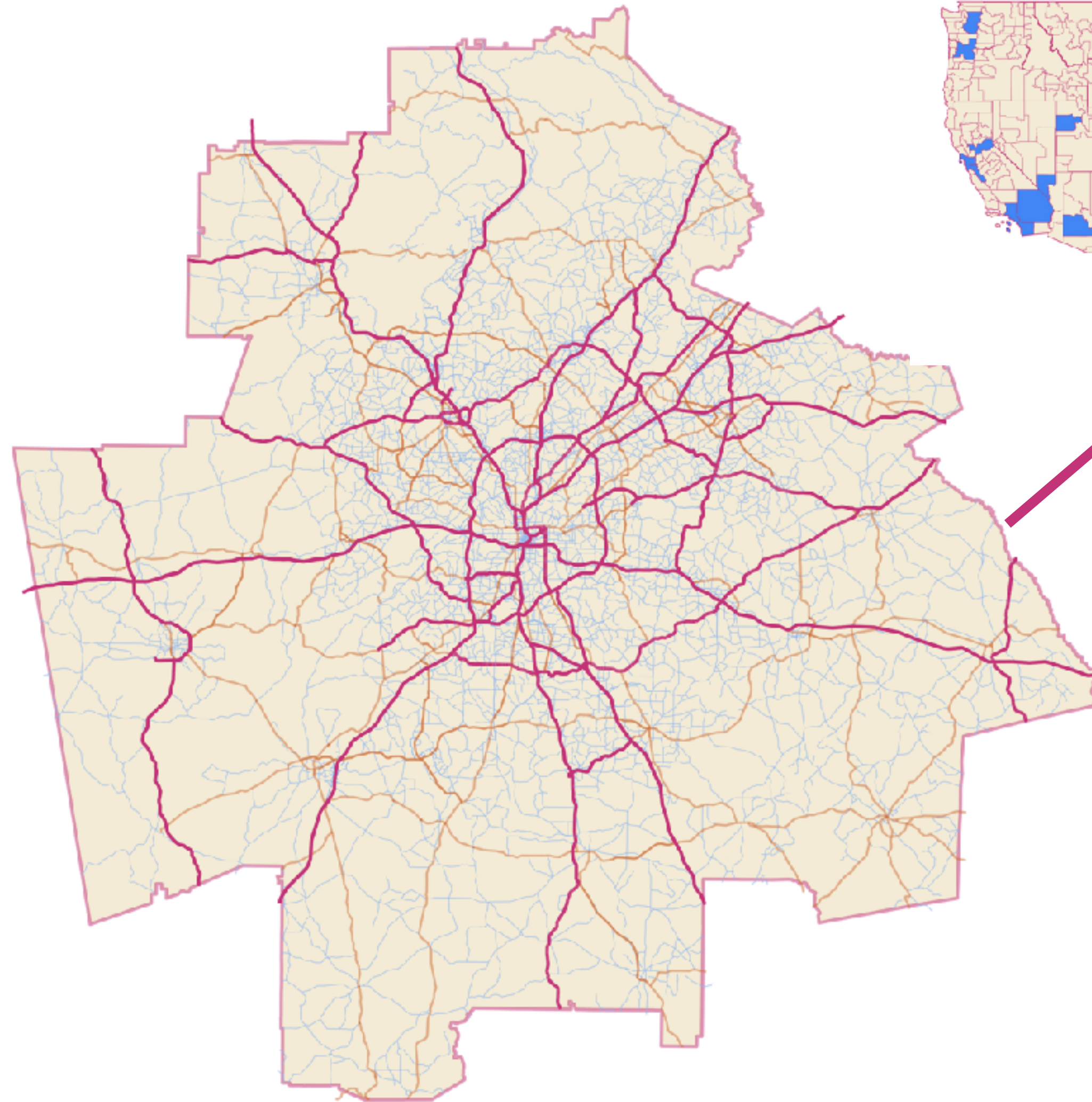


<https://www.theguardian.com/cities/2018/feb/21/roads-nowhere-infrastructure-american-inequality>



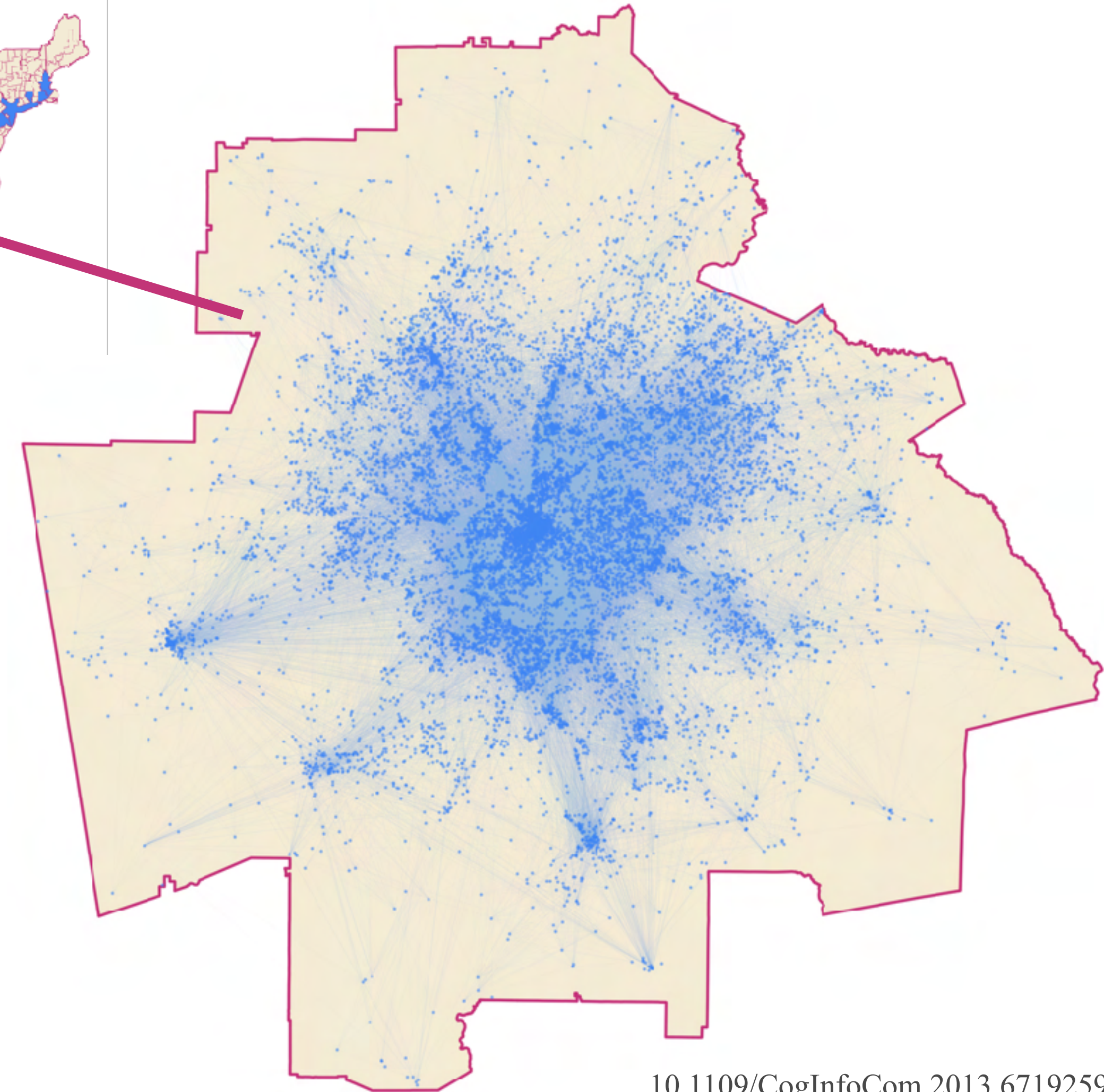
<https://www.pnas.org/doi/10.1073/pnas.2408937122>

Spatial network

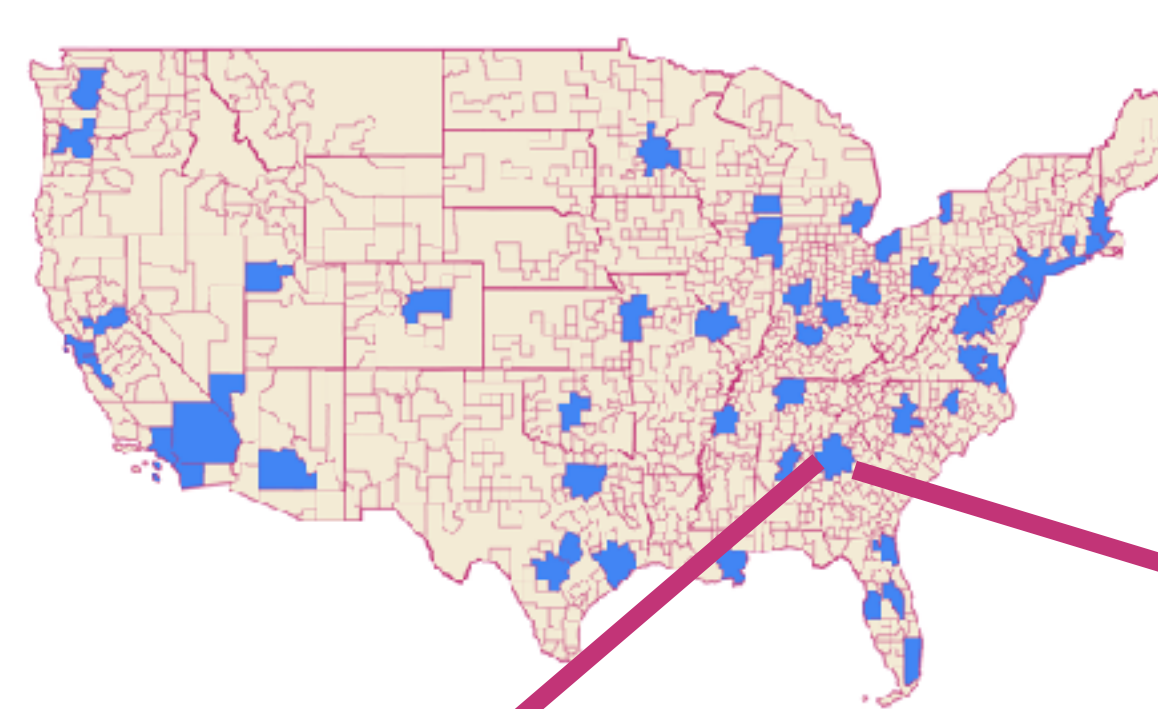


OpenStreetMap

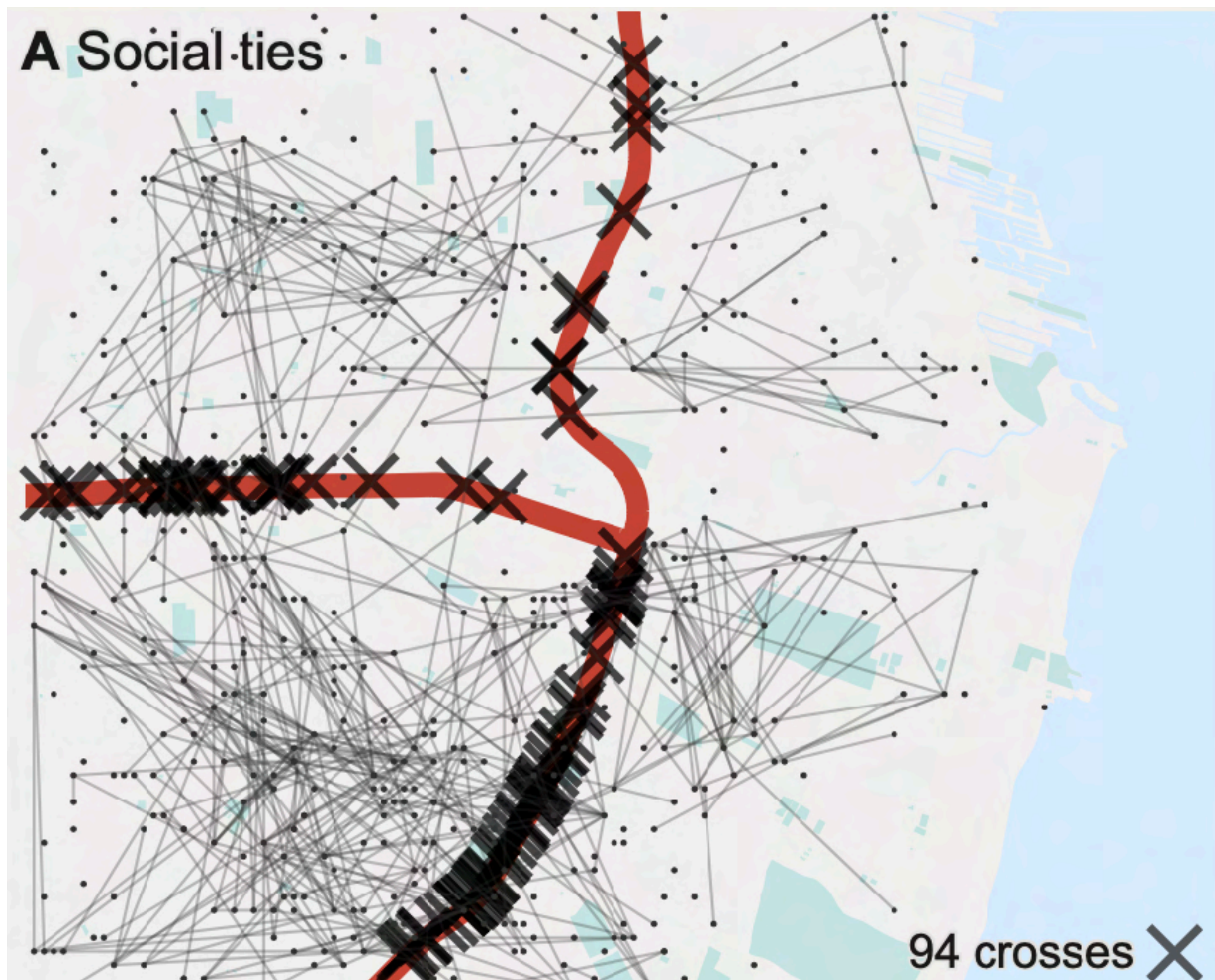
Social network

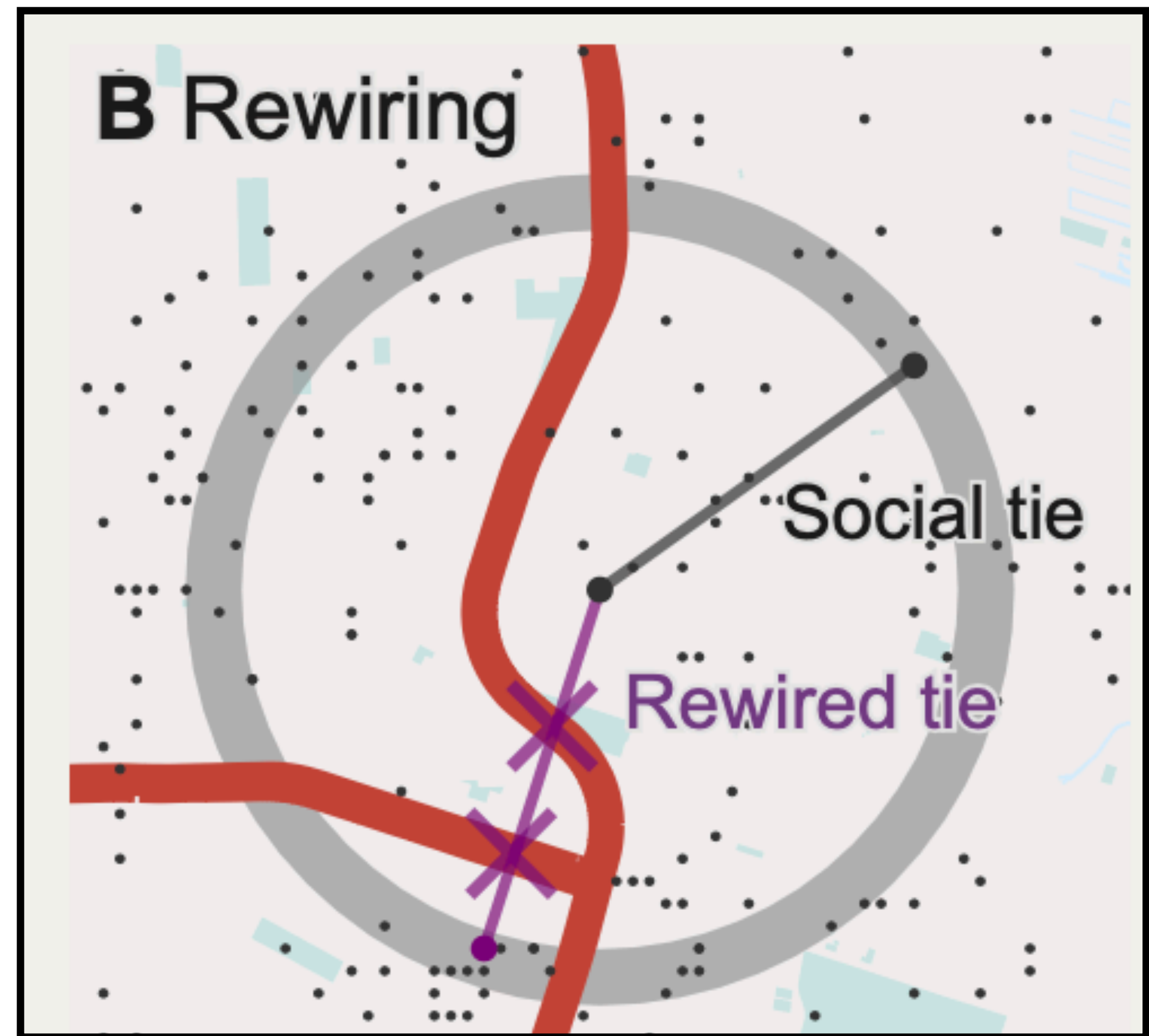
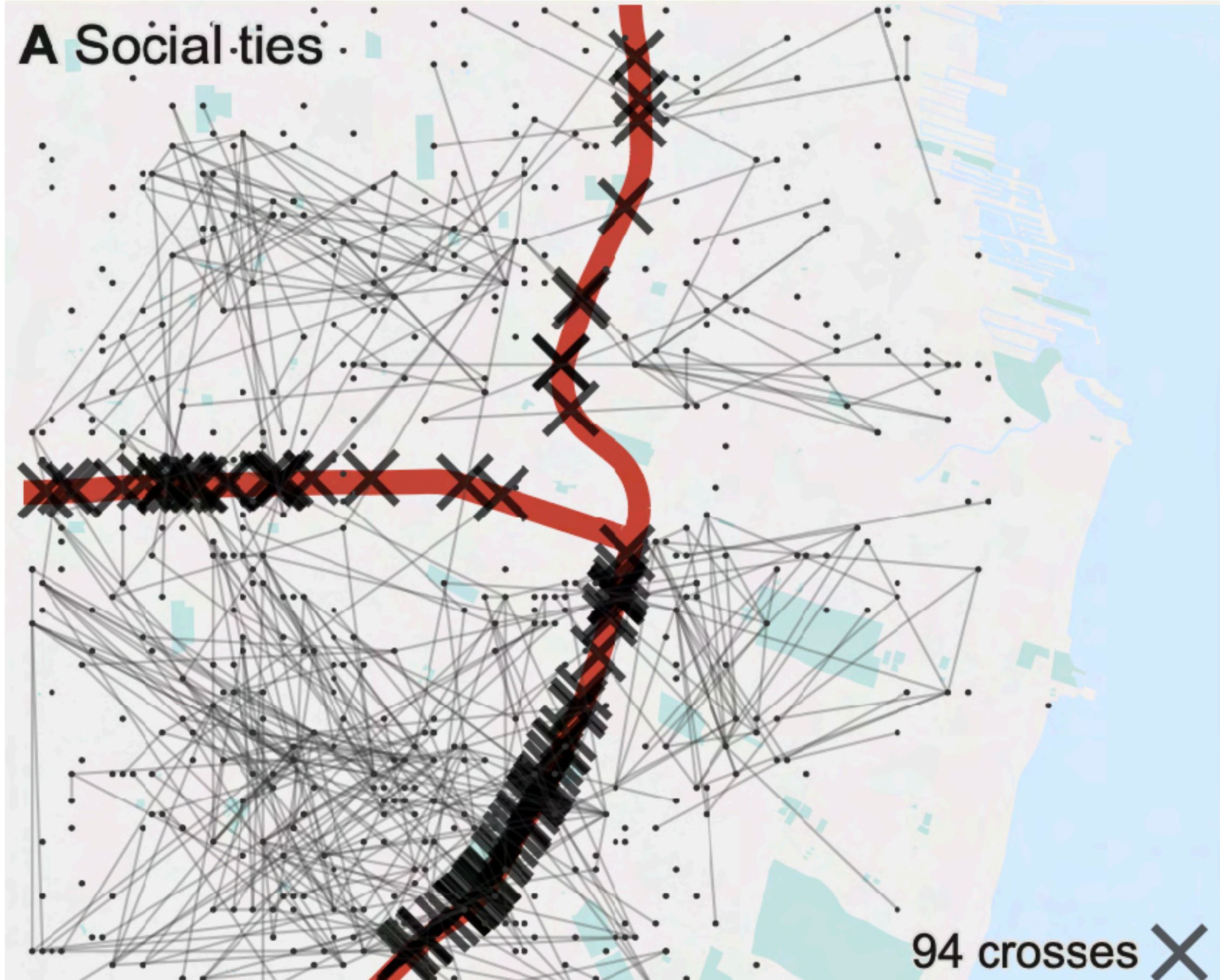


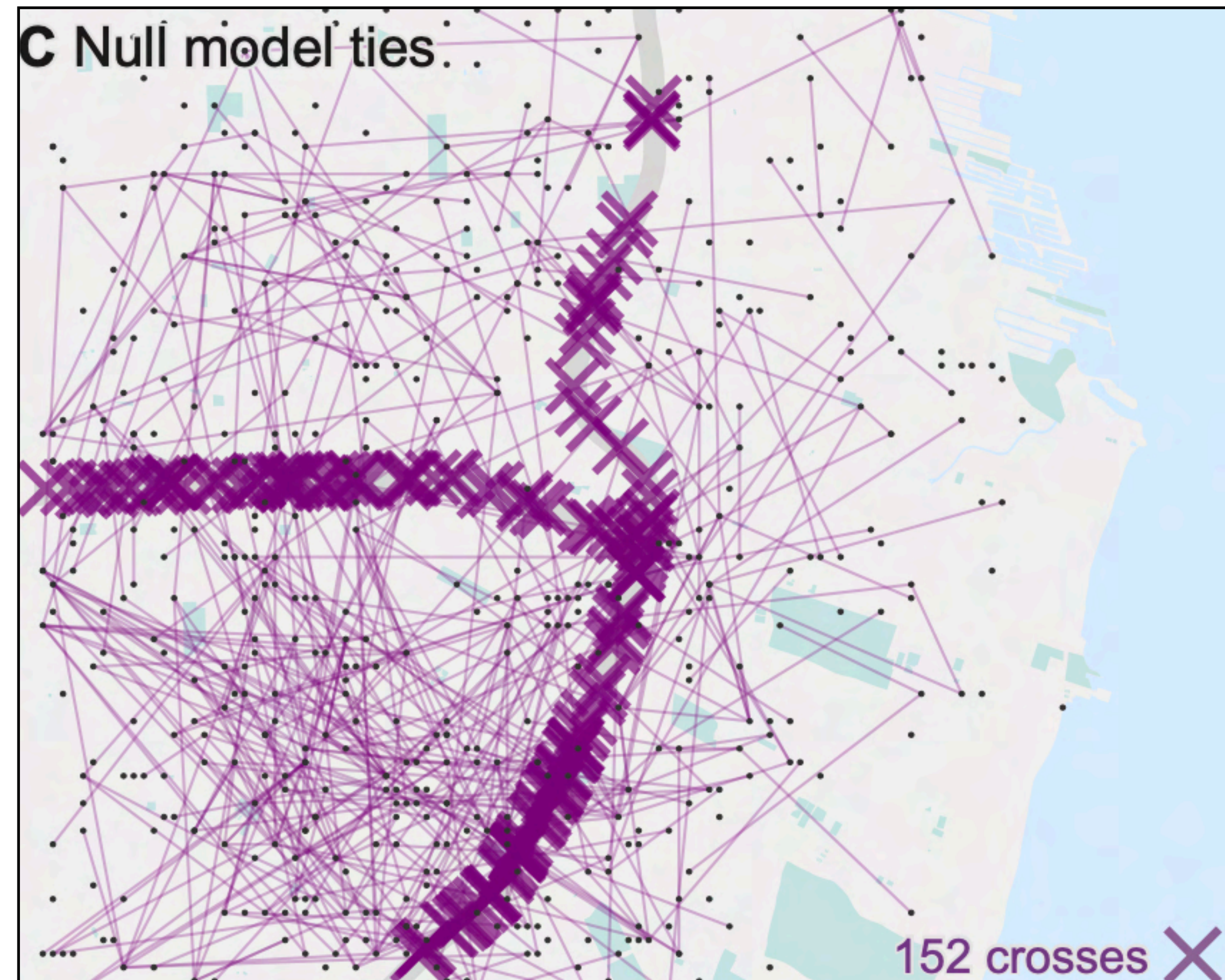
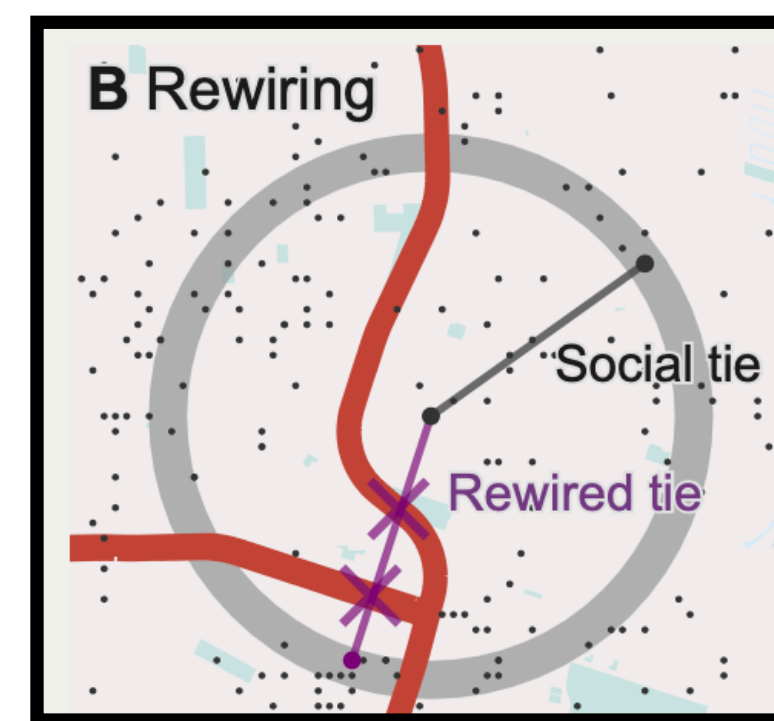
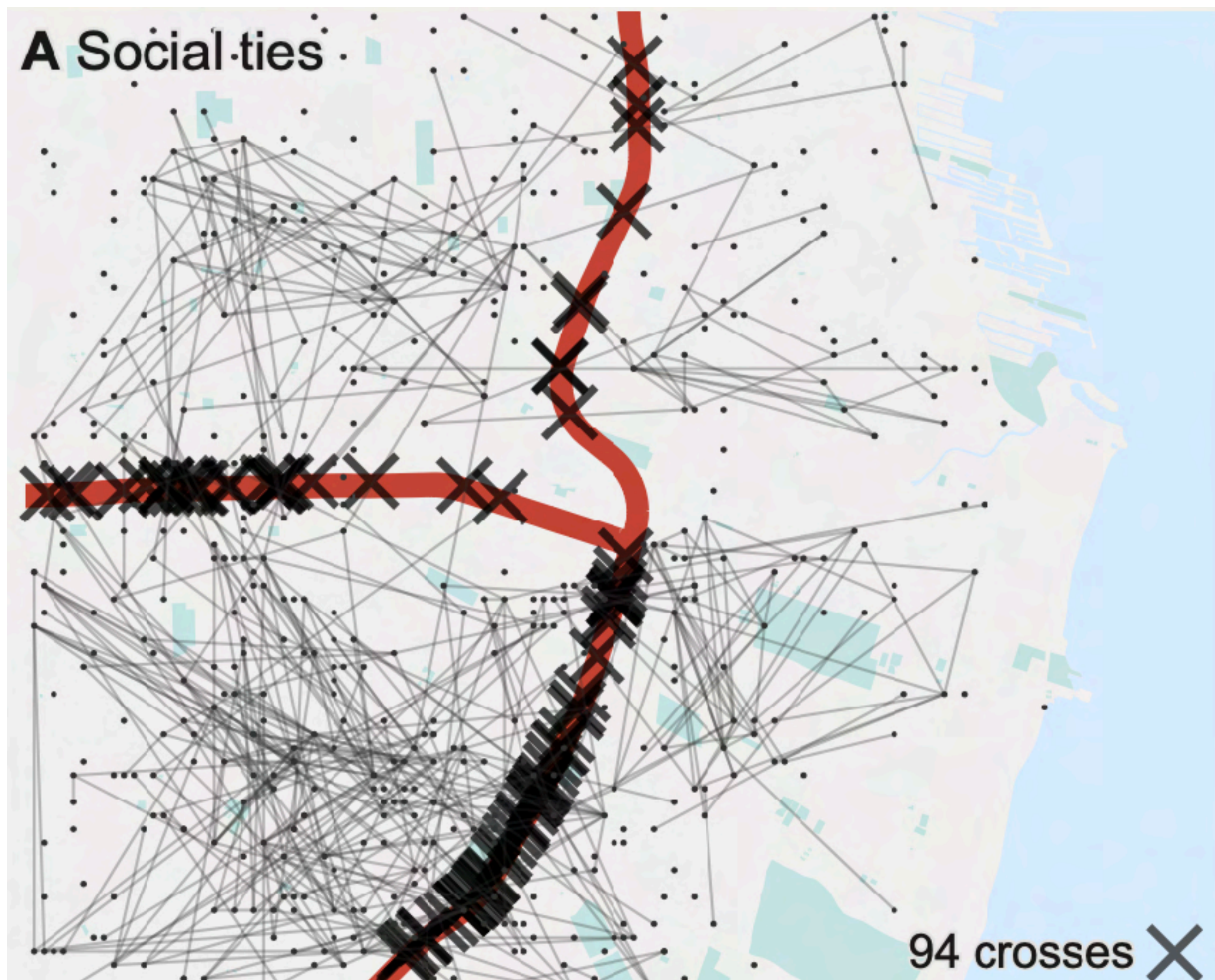
Twitter

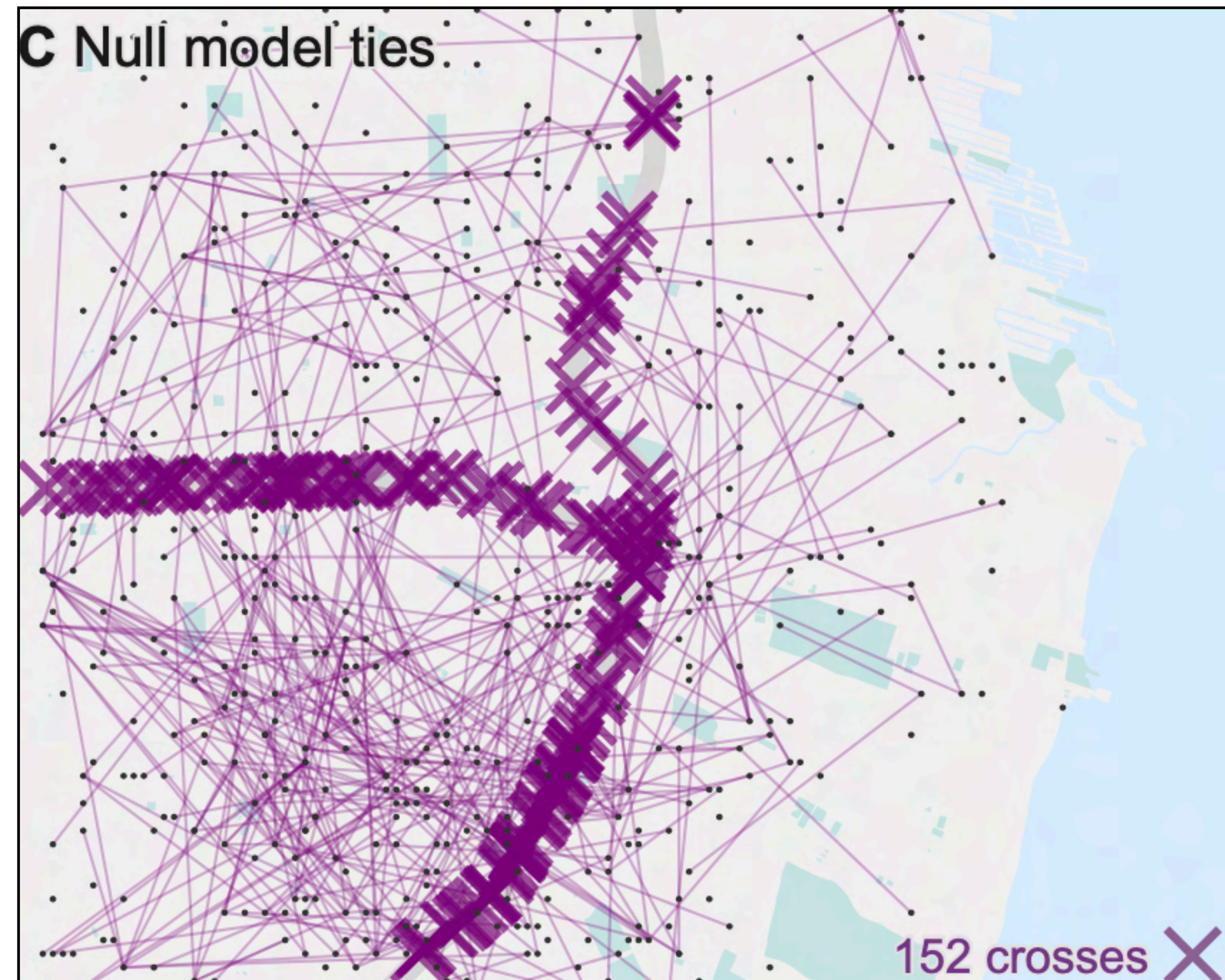
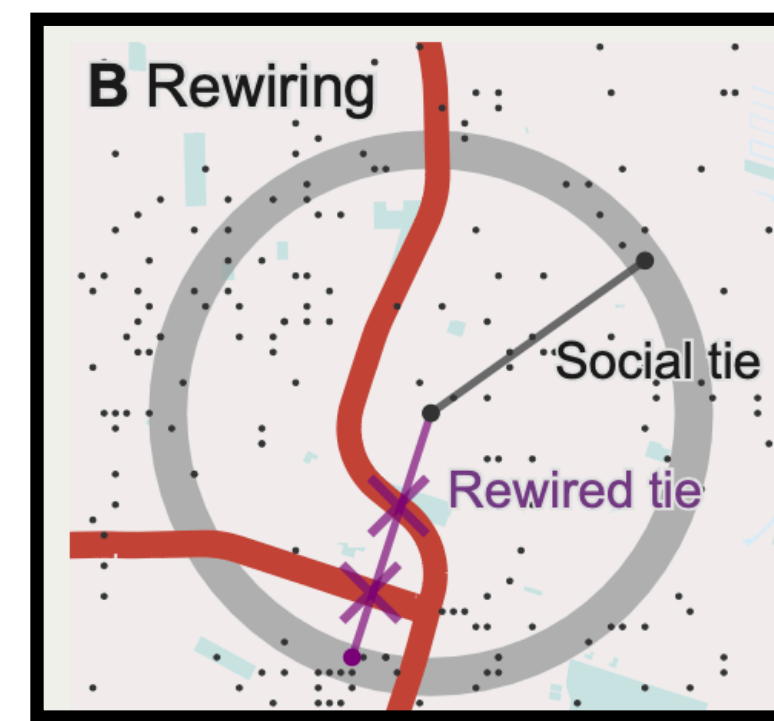
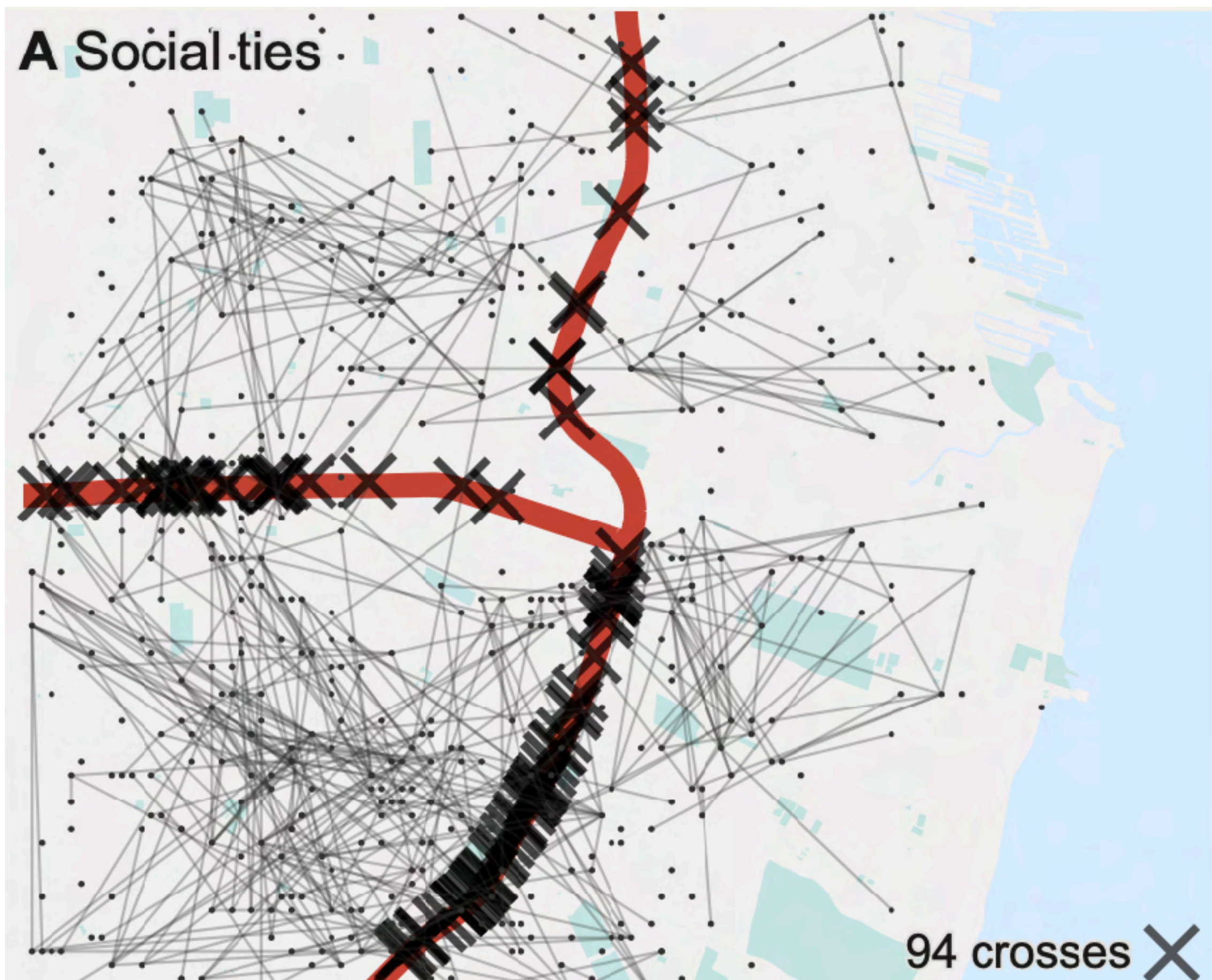


Atlanta, GA





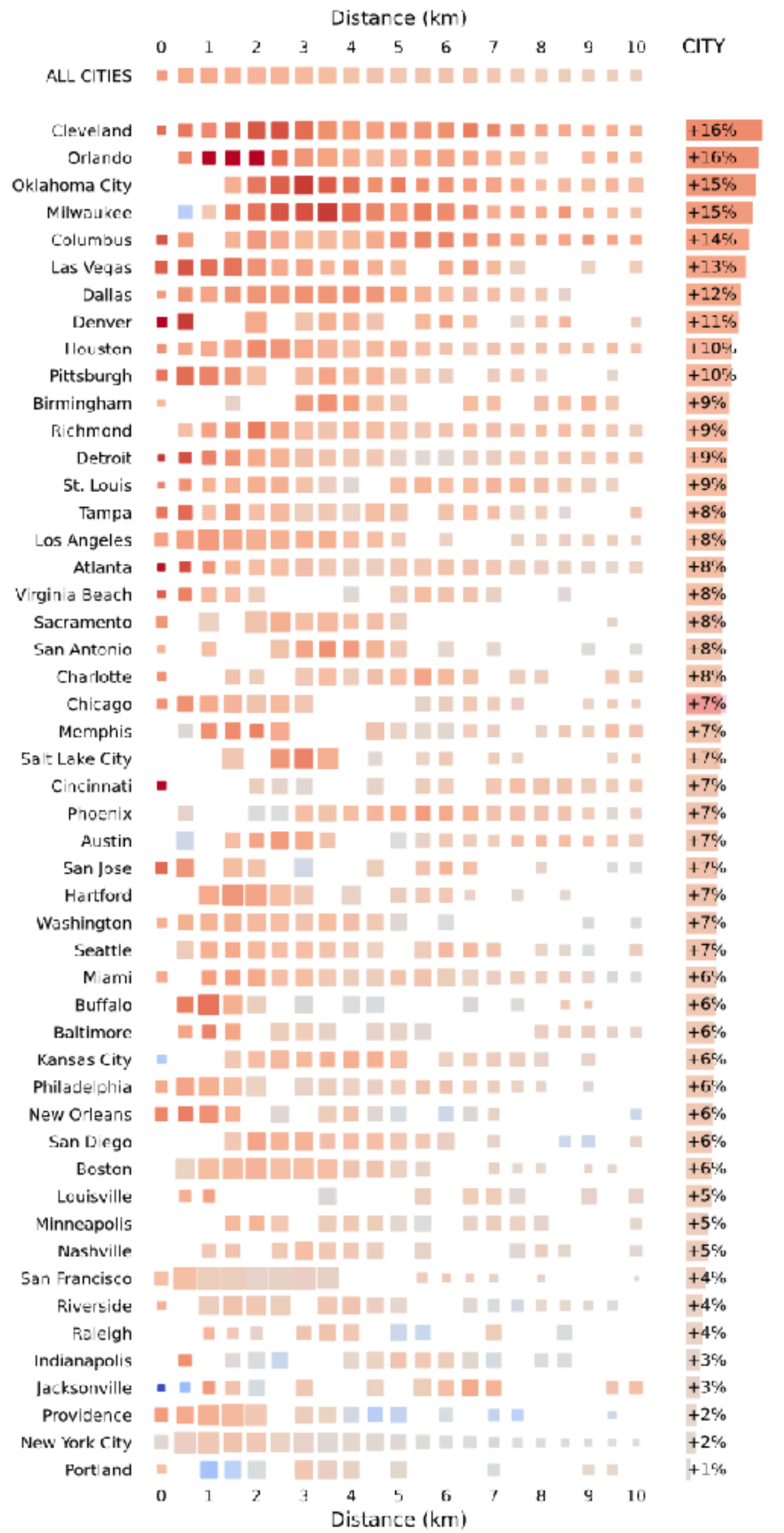
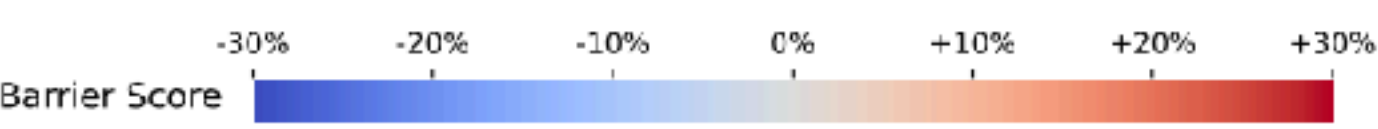


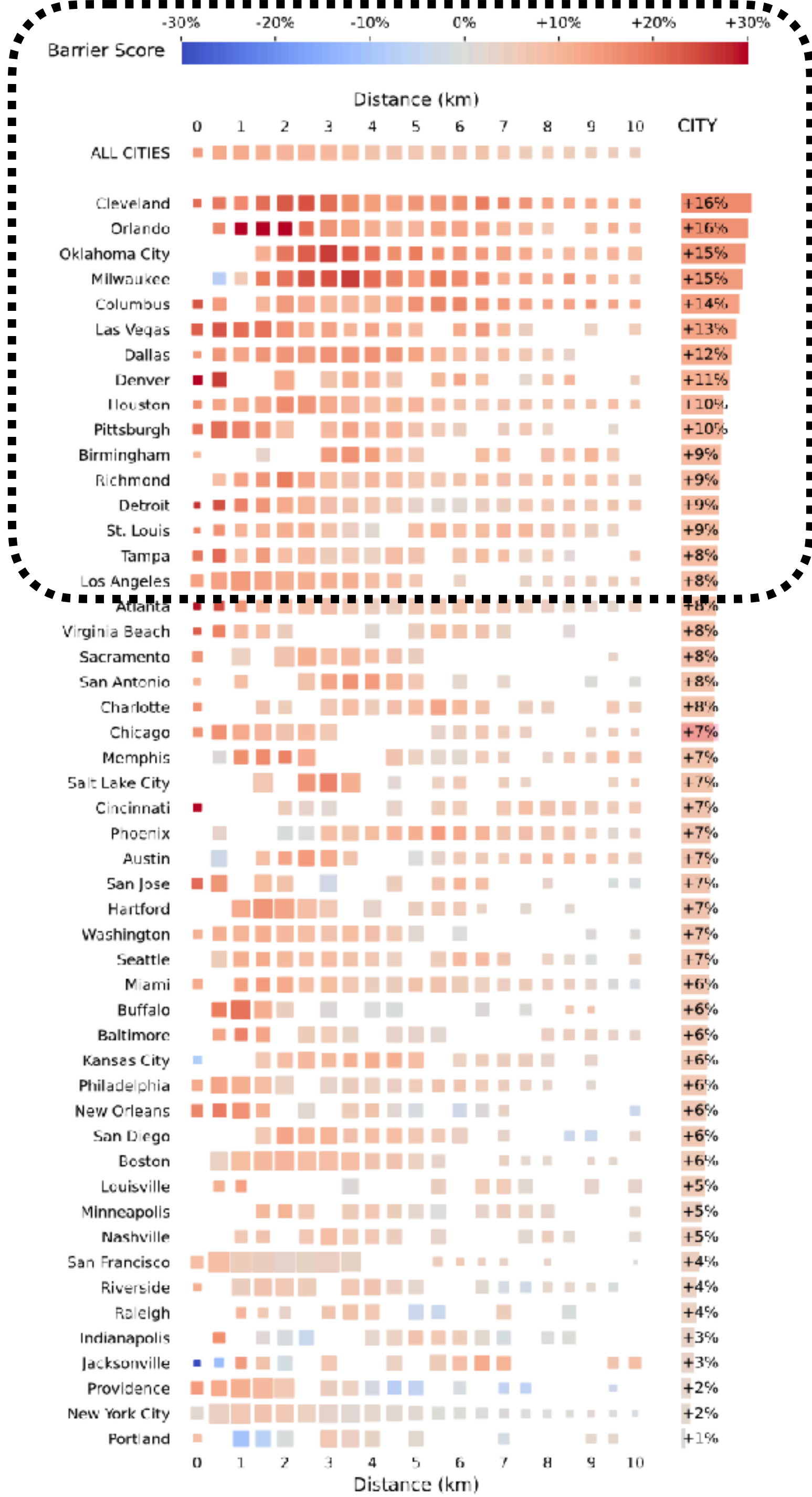


D Barrier Score

$$B_i = \frac{152 - 94}{94}$$

= +62%



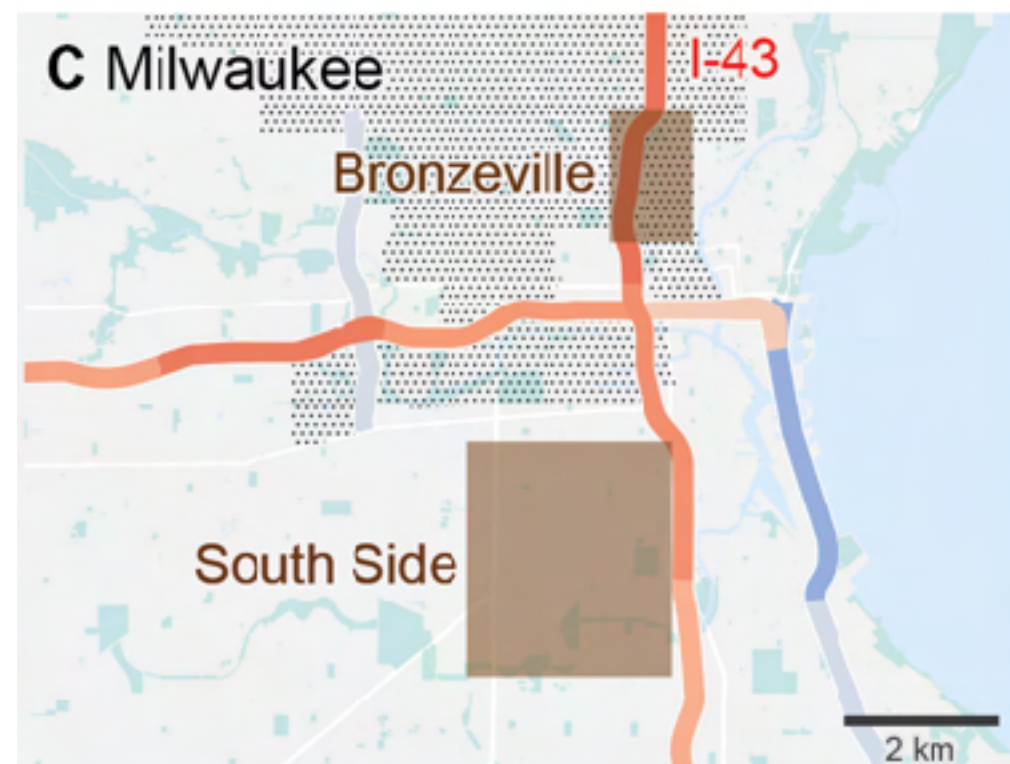
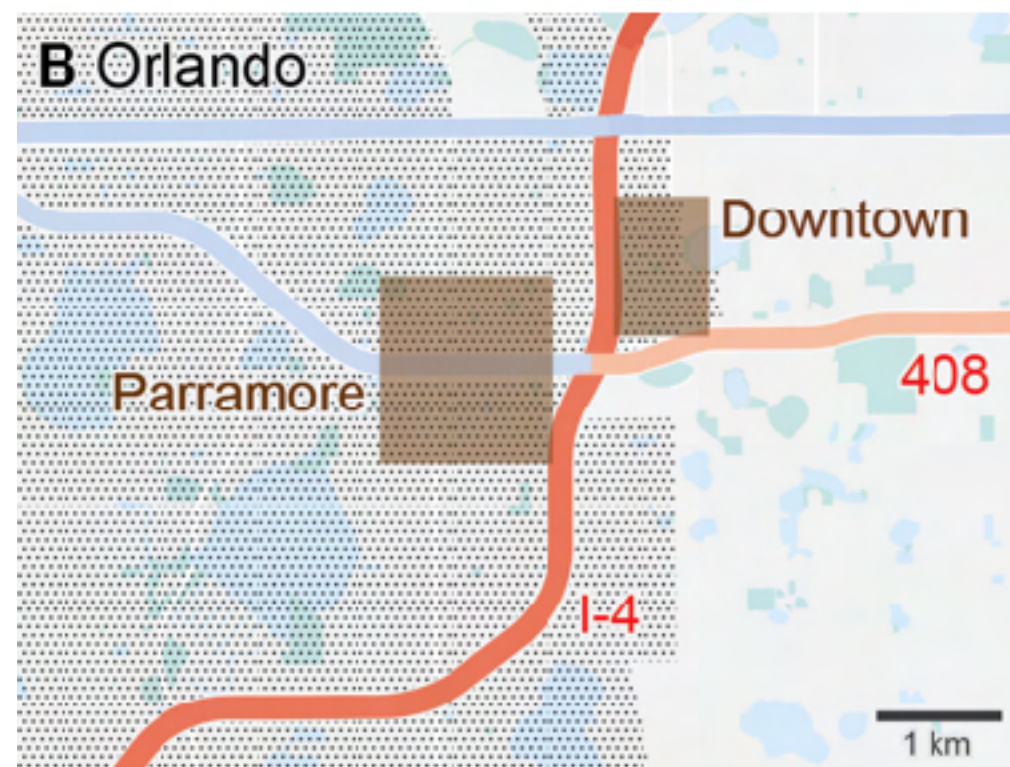


James Baldwin (1963) on urban renewal

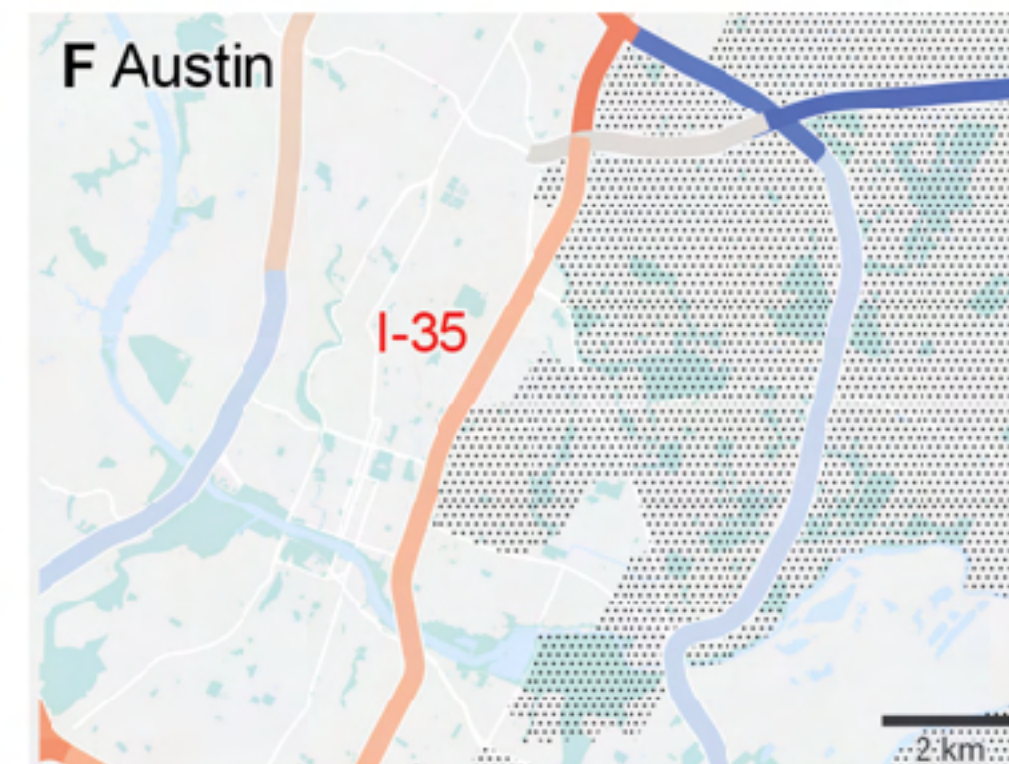
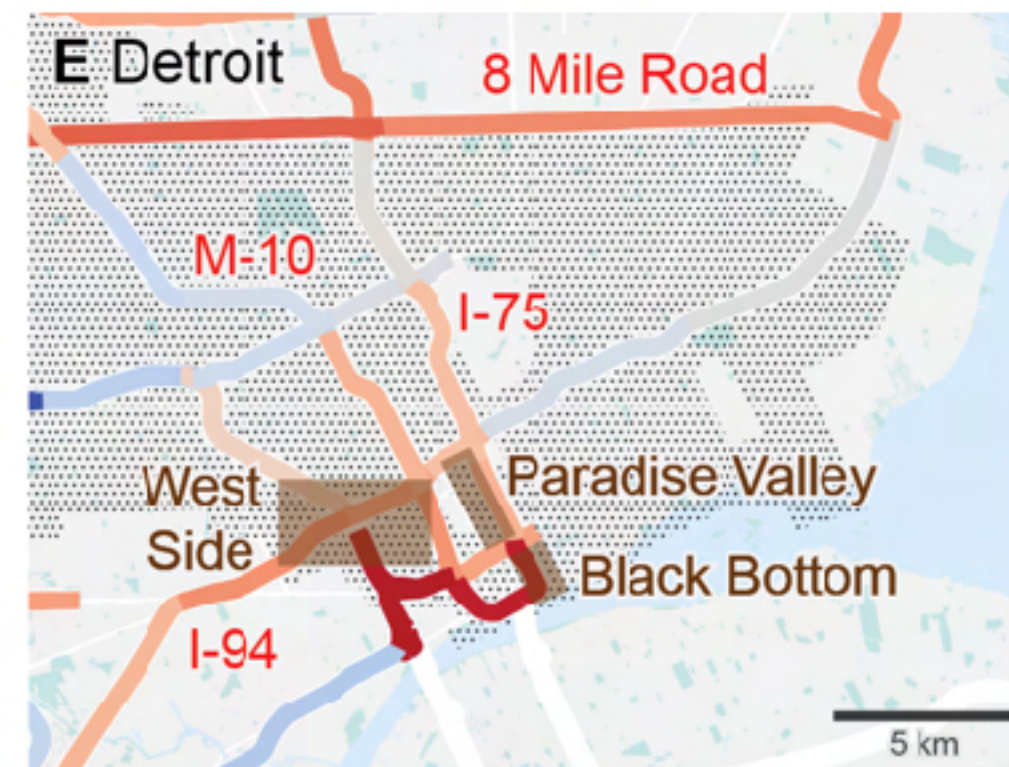
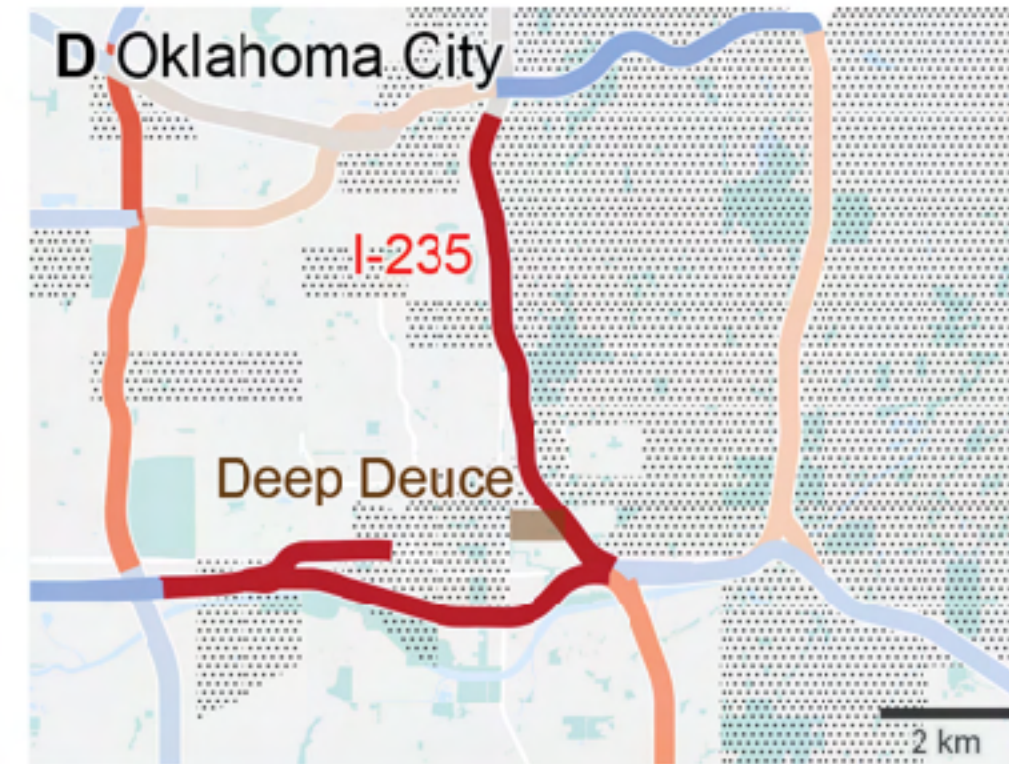


<https://www.youtube.com/watch?v=T8Abhj17kYU>

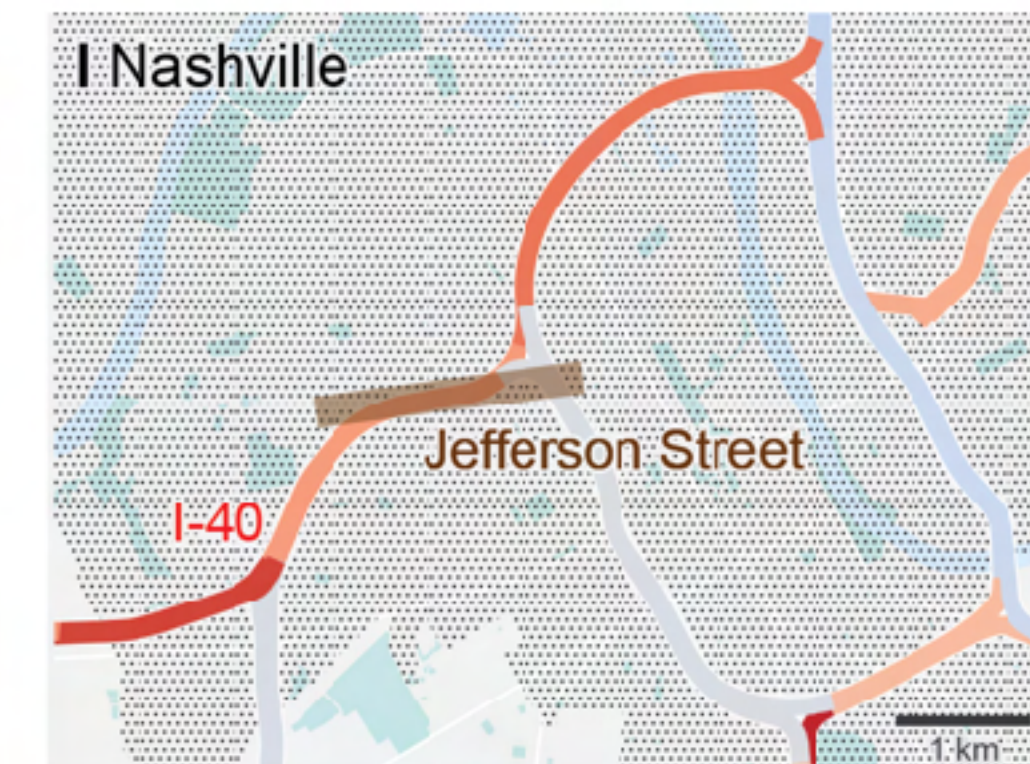
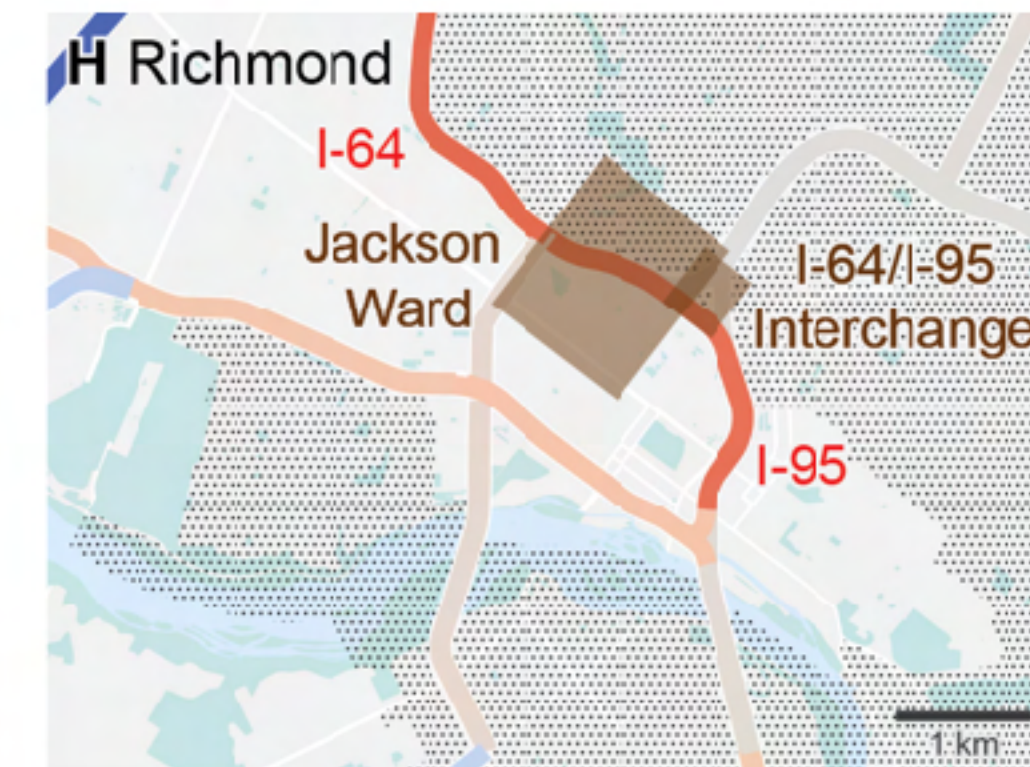
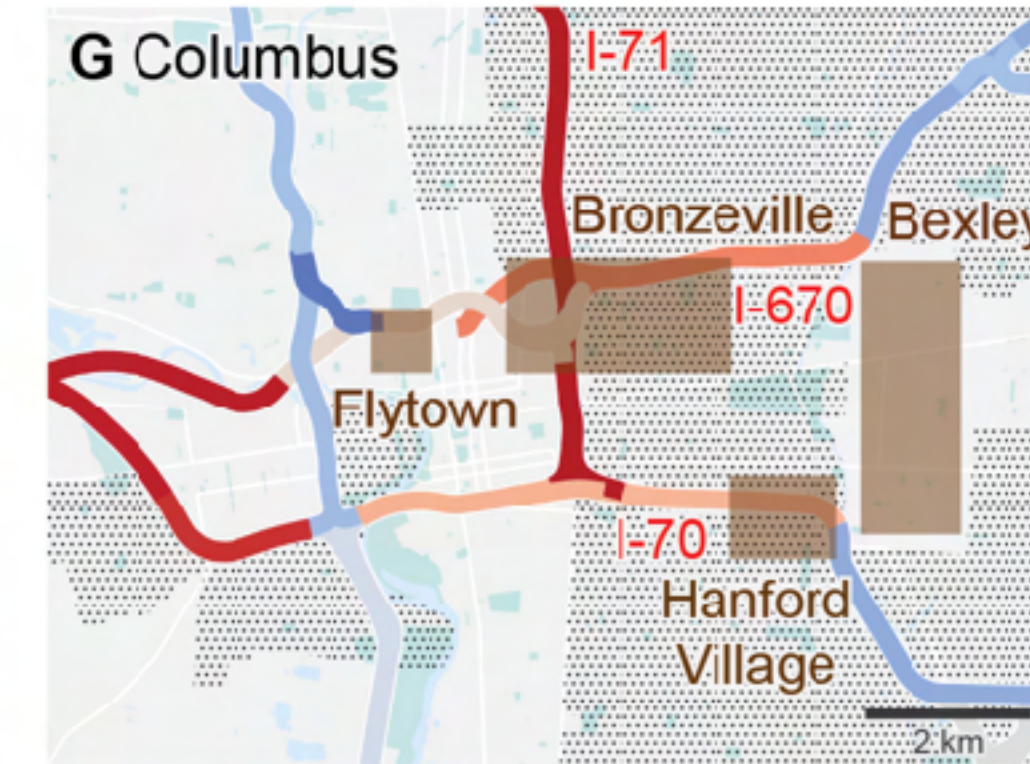
Top 3 Barrier Scores



Interracial Barriers



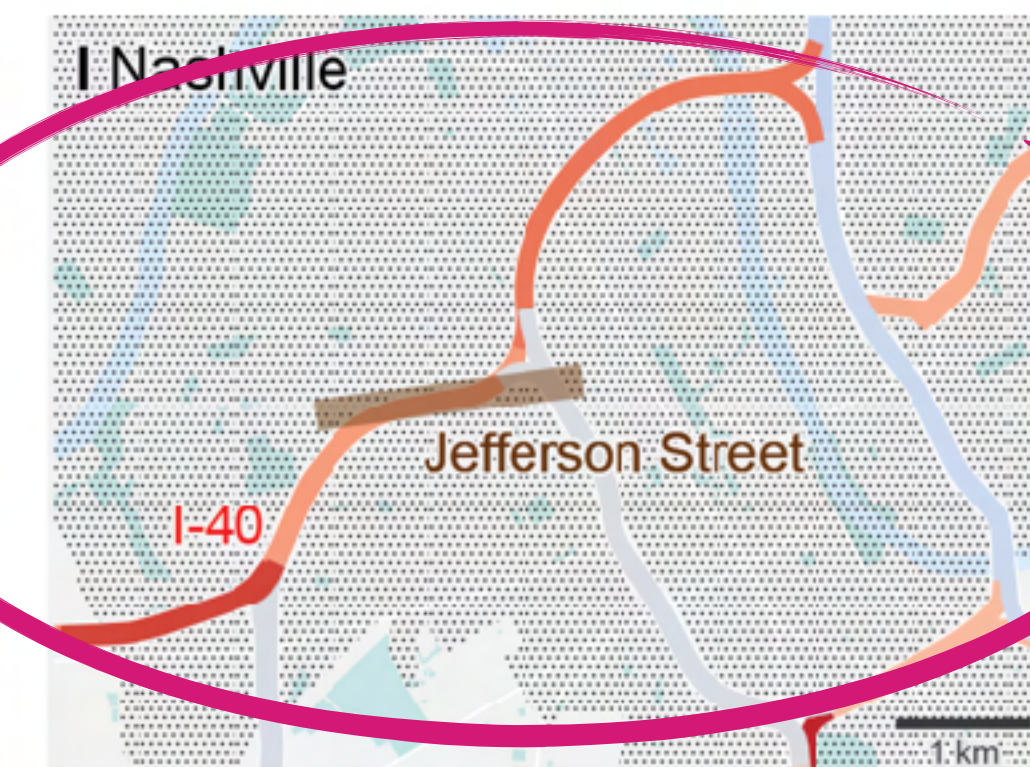
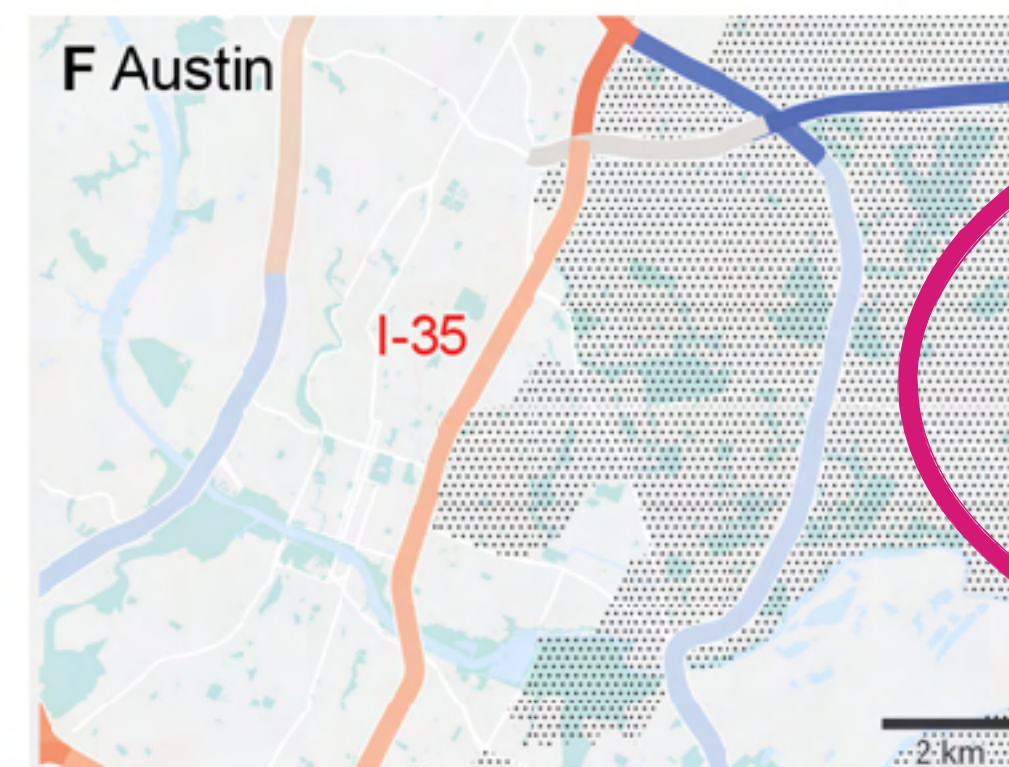
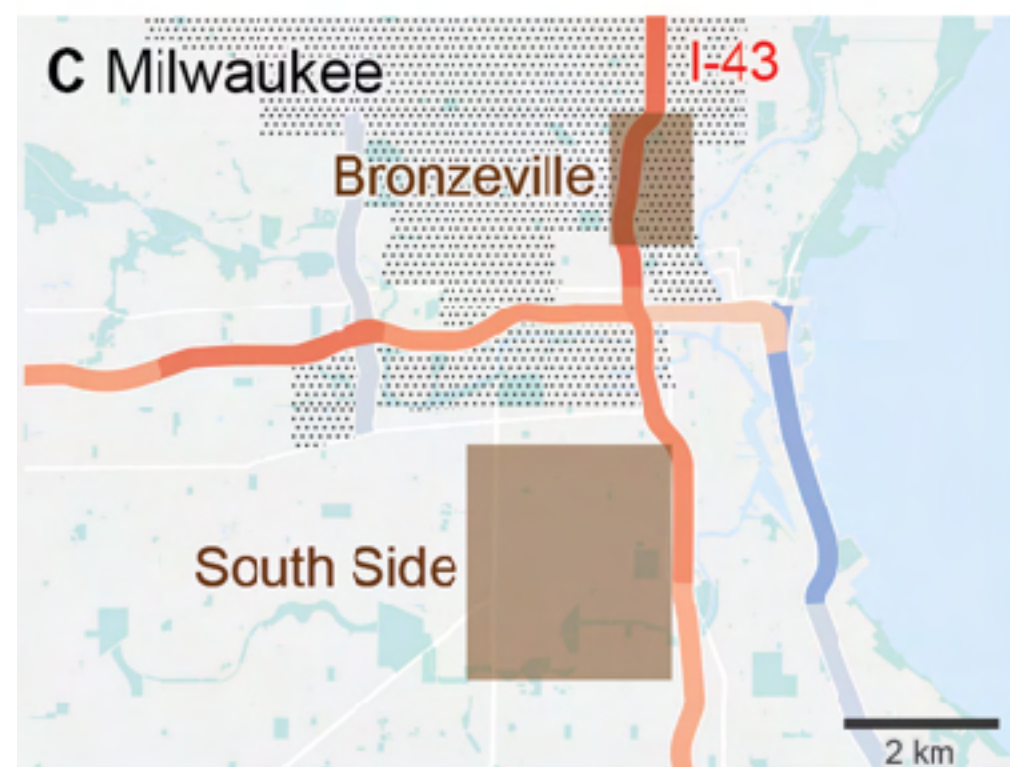
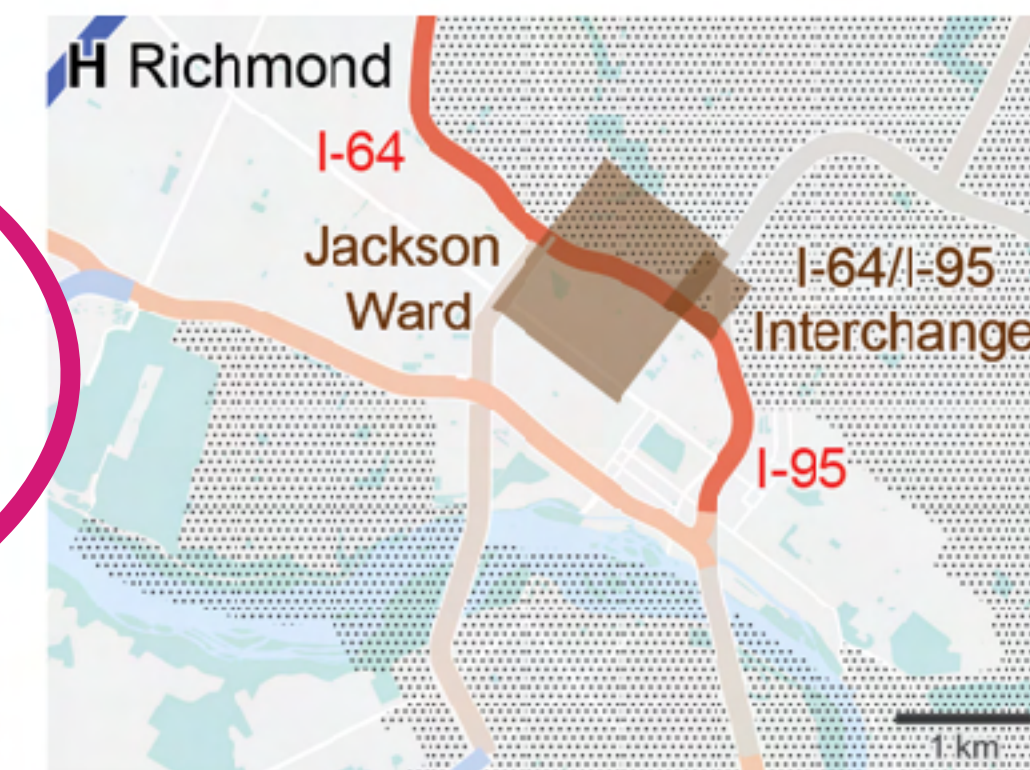
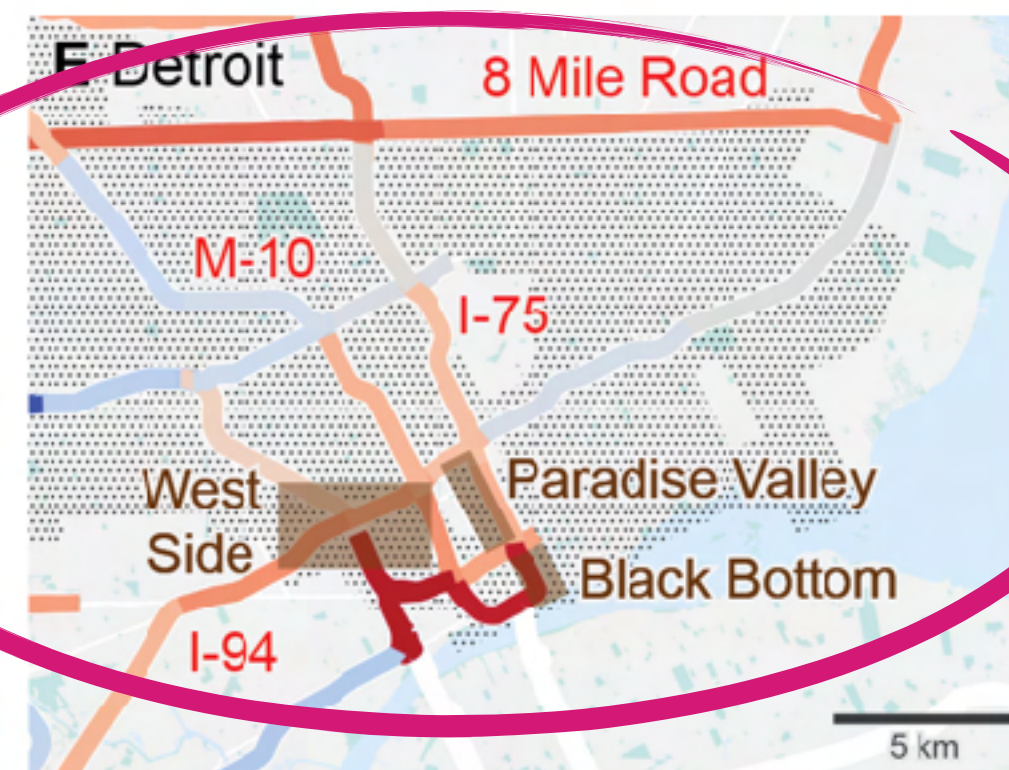
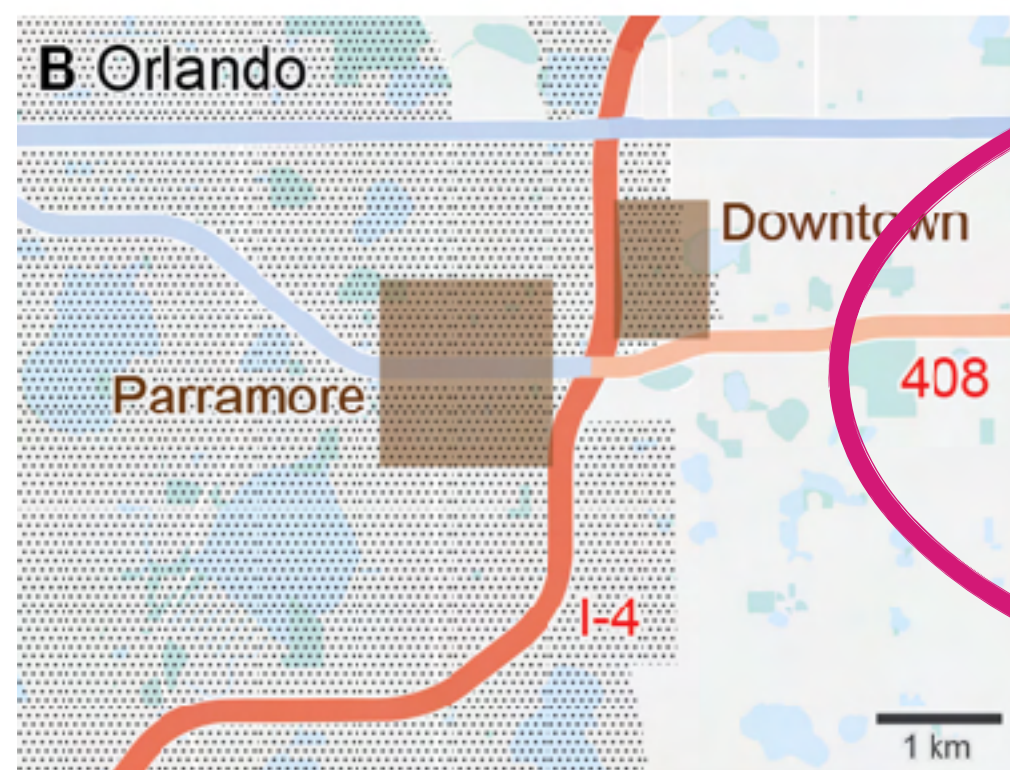
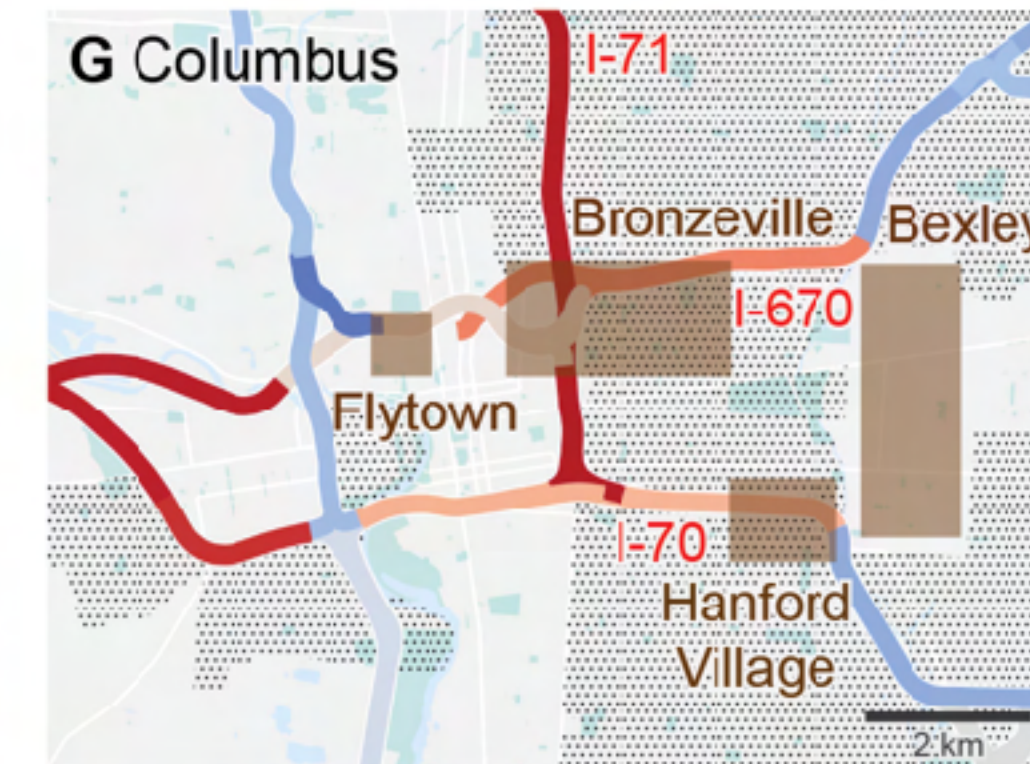
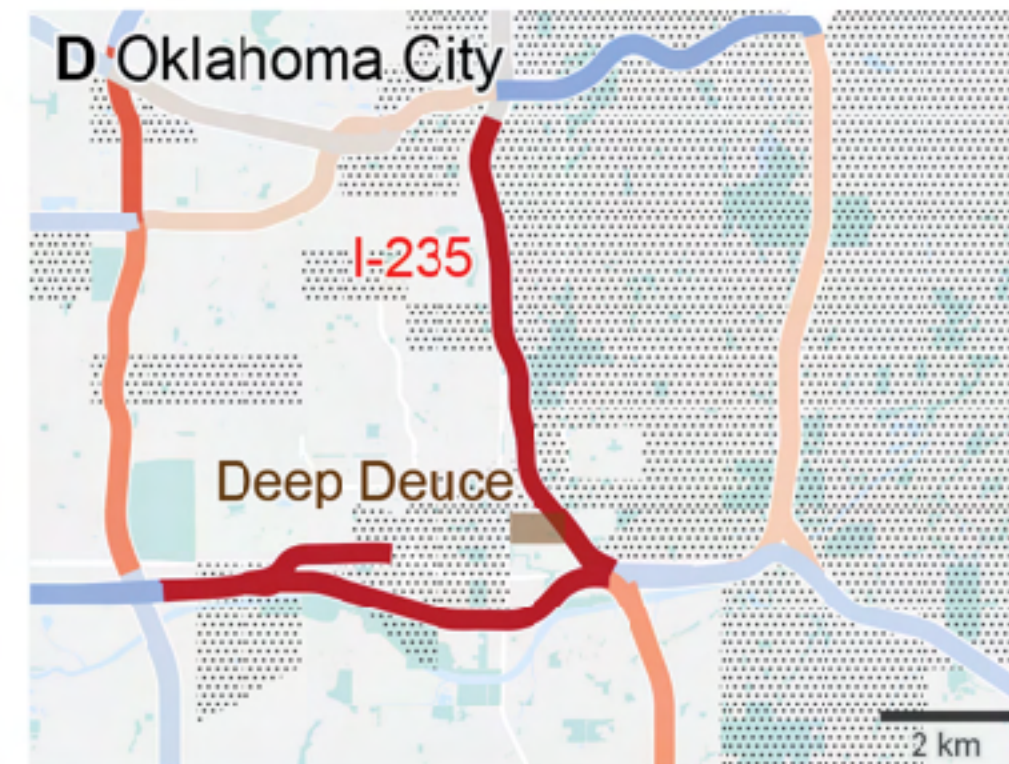
Intraracial Barriers

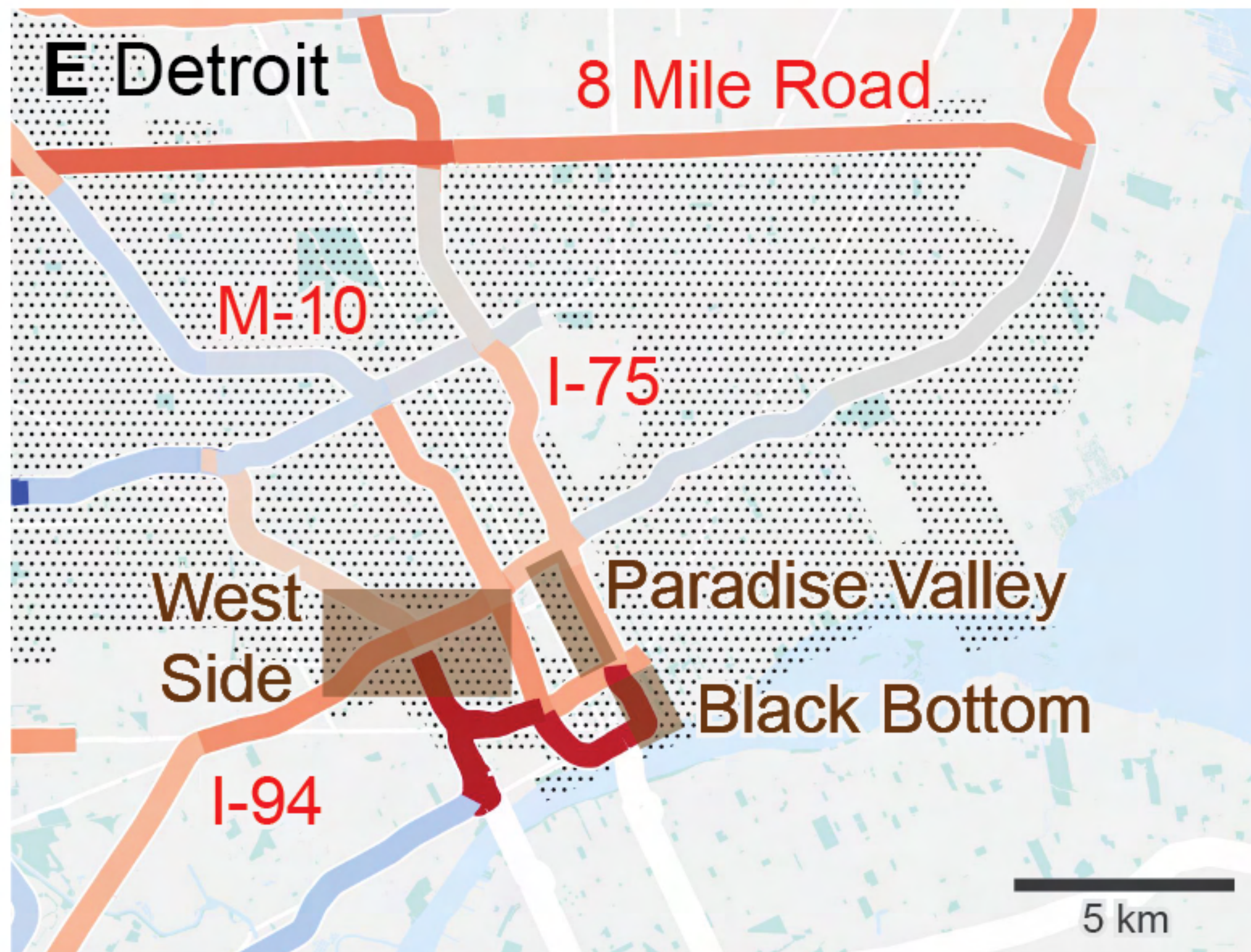


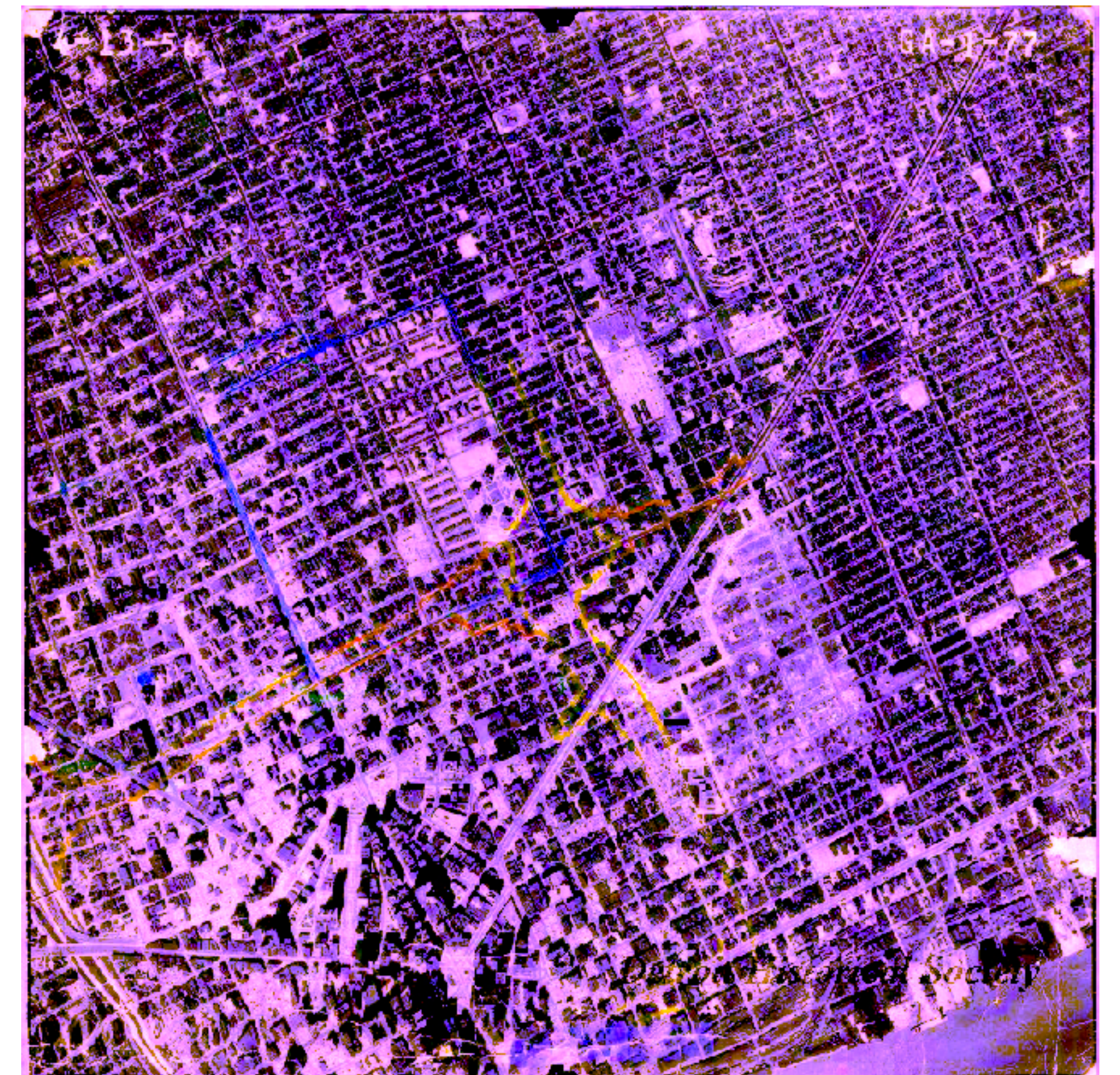
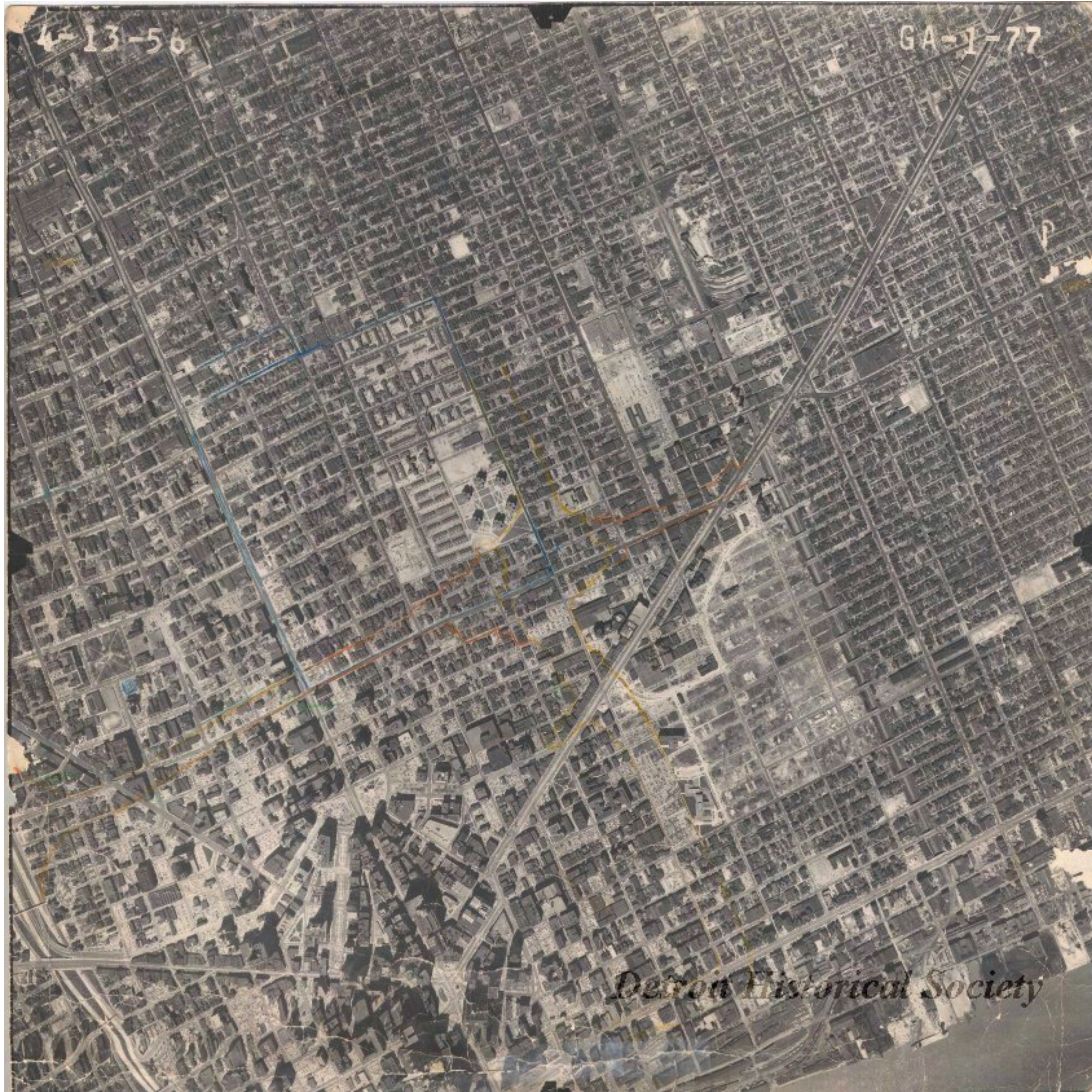
Top 3 Barrier Scores

Interracial Barriers

Intraracial Barriers

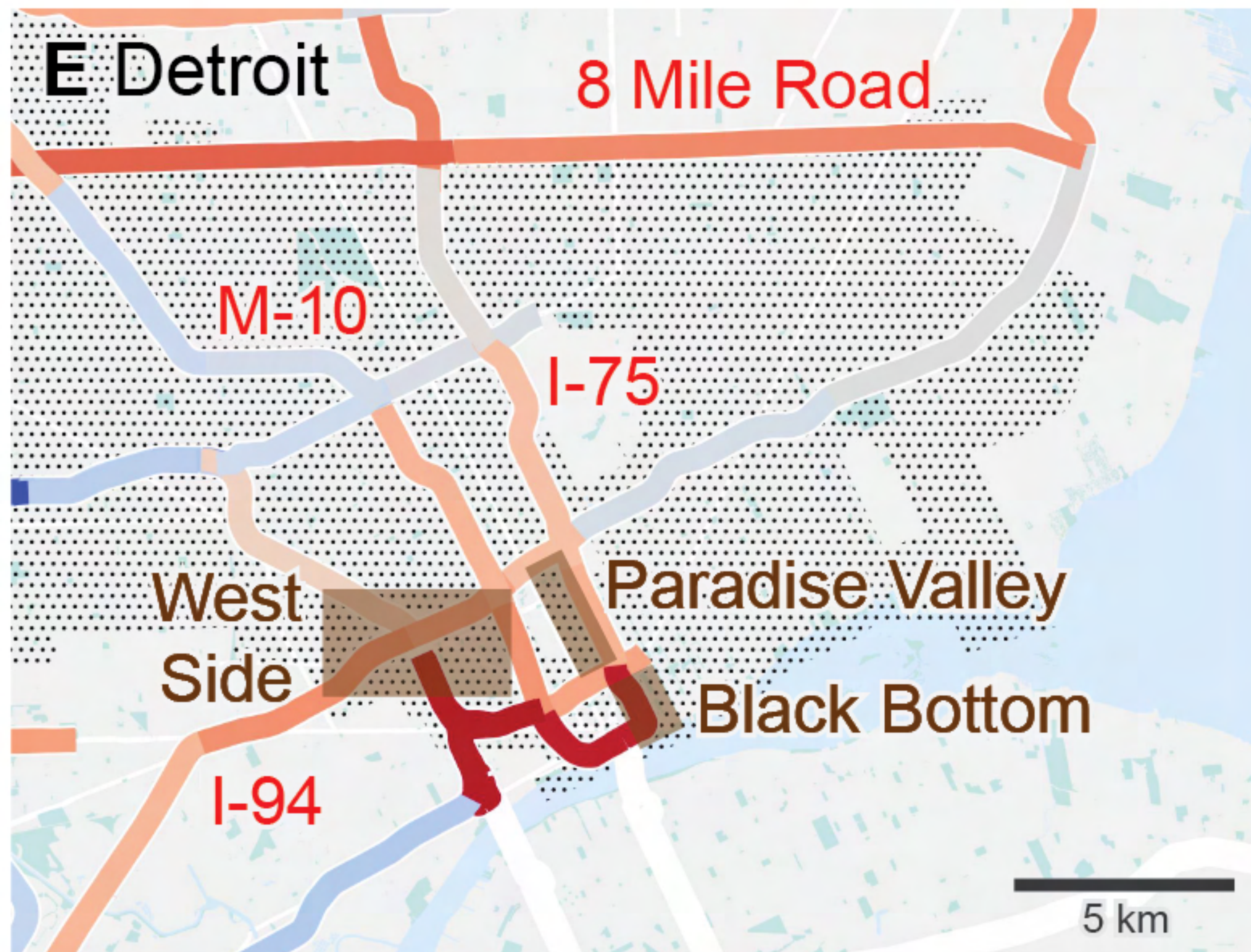






Detroit 1956





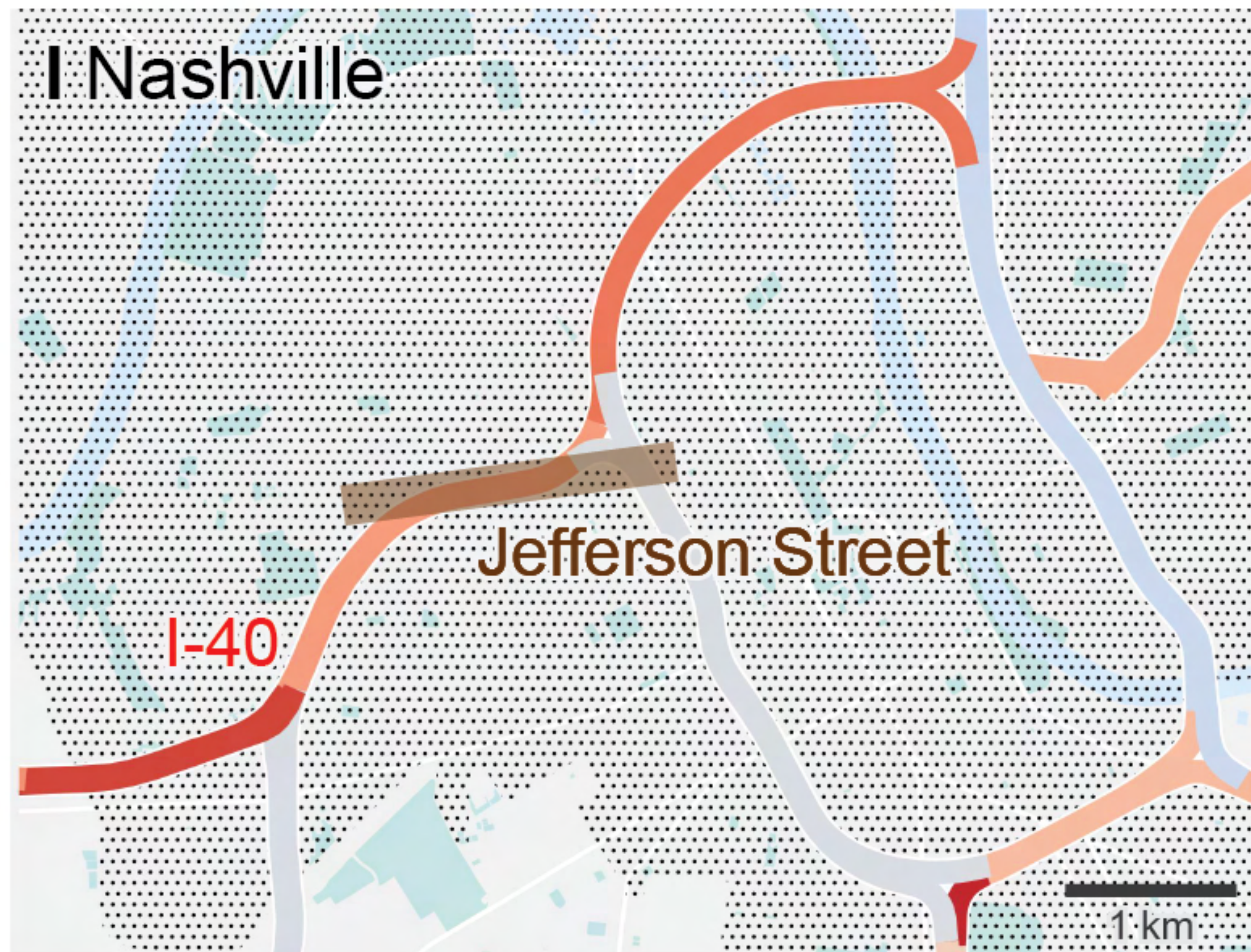


Nashville 1967

<https://eu.tennessean.com/story/news/politics/2021/06/14/nashville-interste-40-cap-project-black-neighborhoods-split-highway/7529846002/>

Nashville I-40 Steering Committee v. Ellington

<https://case-law.vlex.com/vid/nashville-i-40-steering-888302226>



Area of interest

Black, upper quartile

Barrier Score: -30% -15% 0% +15% +30% No data

Reconnecting Communities Program

MARCH 13, 2024

FACT SHEET: President Biden Announces Over \$3 Billion to Reconnect Communities That Have Been Left Behind and Divided by Transportation Infrastructure



BRIEFING ROOM



STATEMENTS AND RELEASES

Portland (OR)

I-5 Rose Quarter Improvement Project



Breaking: House Moves to Rescind \$3.1B for Reconnecting Communities Divided by Highways

The House Transportation Committee wants to slash funding for one of America's most critical equity-focused grant programs — unless advocates speak out and get them to reverse course.



By **Kea Wilson**

4:13 PM GMT-4 on April 29, 2025

Note: America Walks is collecting signatures from organizations who oppose this proposal. Sign on here. Private individuals who want to contact their representatives to urge them to preserve this funding can do so through with this tool from the National Campaign for Justice.

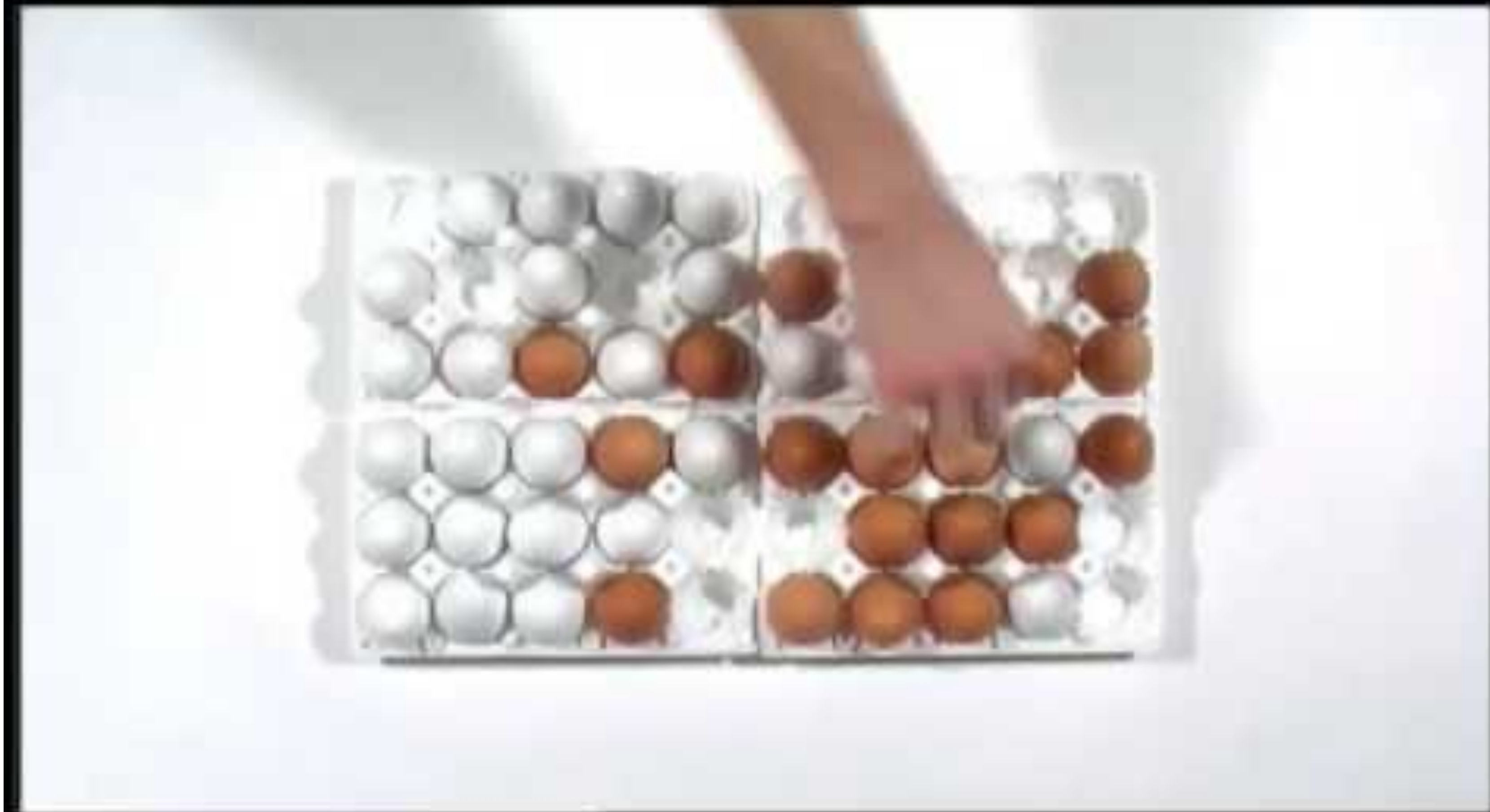
Act II: You Don't Have To Live Next To Me

Towards Demobilizing Individualistic Bias in Computational Approaches to Urban Segregation

Anastassia Vybornova & Trivik Verma

Artwork by Namrata Narendra

<https://arxiv.org/abs/2505.01830>



<https://youtube.com/watch?v=JjfihtGefxk>

Fifty years after the Schelling's Models of Segregation: Bibliometric analysis of the legacy of Schelling and the future directions of segregation research

Rūta Ubarevičienė^{a,b,c,*}, Maarten van Ham^a, Tiit Tammaru^{b,d}

^a *Department of Urbanism, Delft University of Technology, Delft, the Netherlands*

^b *Department of Geography, University of Tartu, Tartu, Estonia*

^c *Department of Regional and Urban Studies, Institute of Sociology, Lithuanian Centre for Social Sciences, Vilnius, Lithuania*

^d *Estonian Academy of Sciences, Estonia*

It has been more than 50 years since Thomas Schelling (1921–2016) developed theoretical models explaining the dynamics of residential segregation. Today, the Schelling models of residential segregation have become one of the most cited models in the urban research literature.

explains clustering and separating, the most important issue in the Schelling model, from a social perspective, is how choices play out in the social fabric and lead to segregated residential patterns. We show here that there are well articulated mathematical explanations for social segregation, that simulation compared with having utility zero. Schelling used simple simulations based on such hypothetical preference schedules to show that the adjustments of individual households responding to changes in composition on the checkerboard invariably lead to complete segregation (3). New survey evidence for

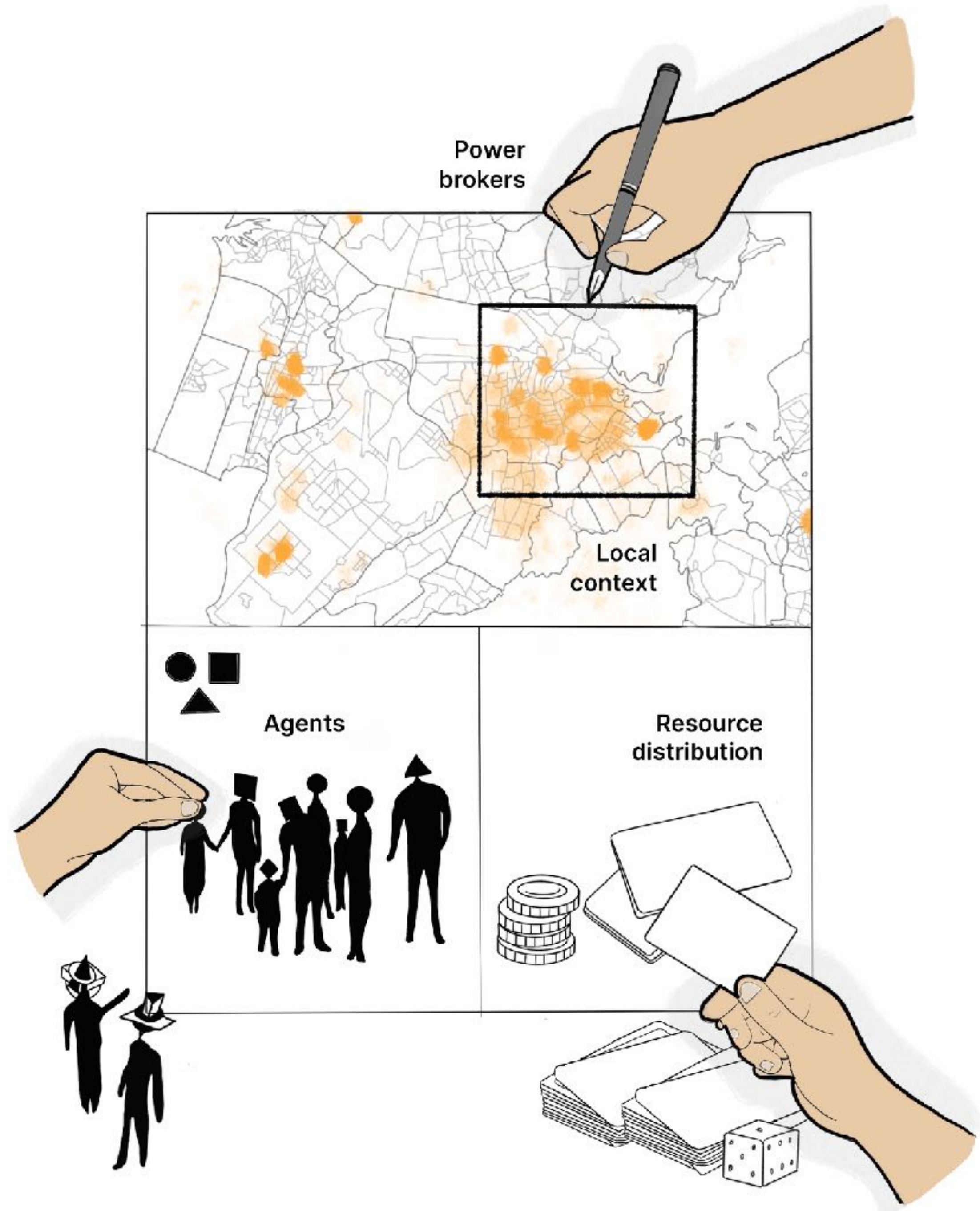
Clark & Fossett (2008) <https://www.jstor.org/stable/25461379>

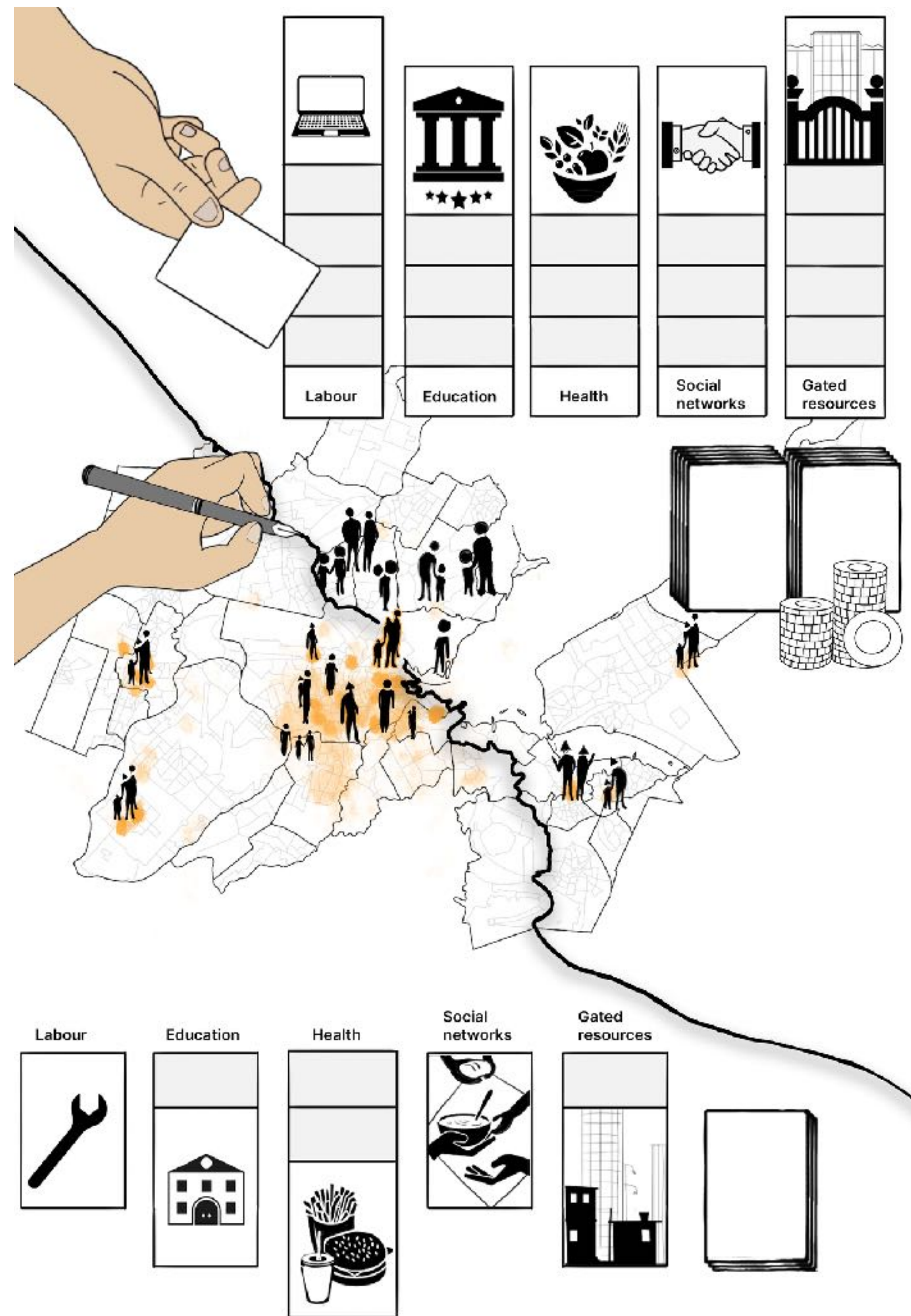
One of the most striking examples of this is residential segregation (Sakoda 1971; Schelling 1971), where subtle behavioural responses at a microlevel are well known to have the potential to lead to severe consequences such as ethnic segregation at the macrolevel. A good understanding of individual

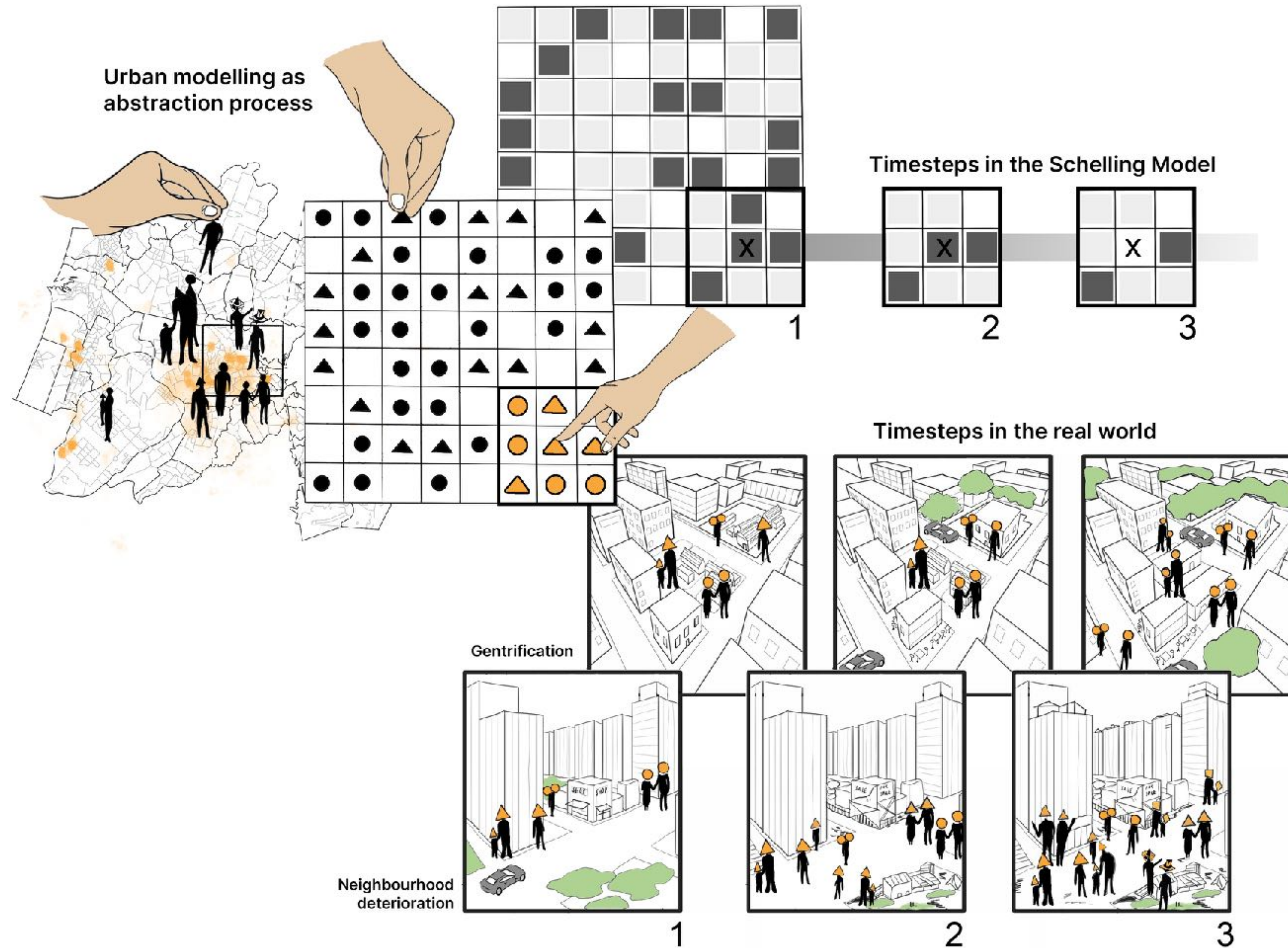
Liebe et al. (2023) doi:10.1177/00491241231186657

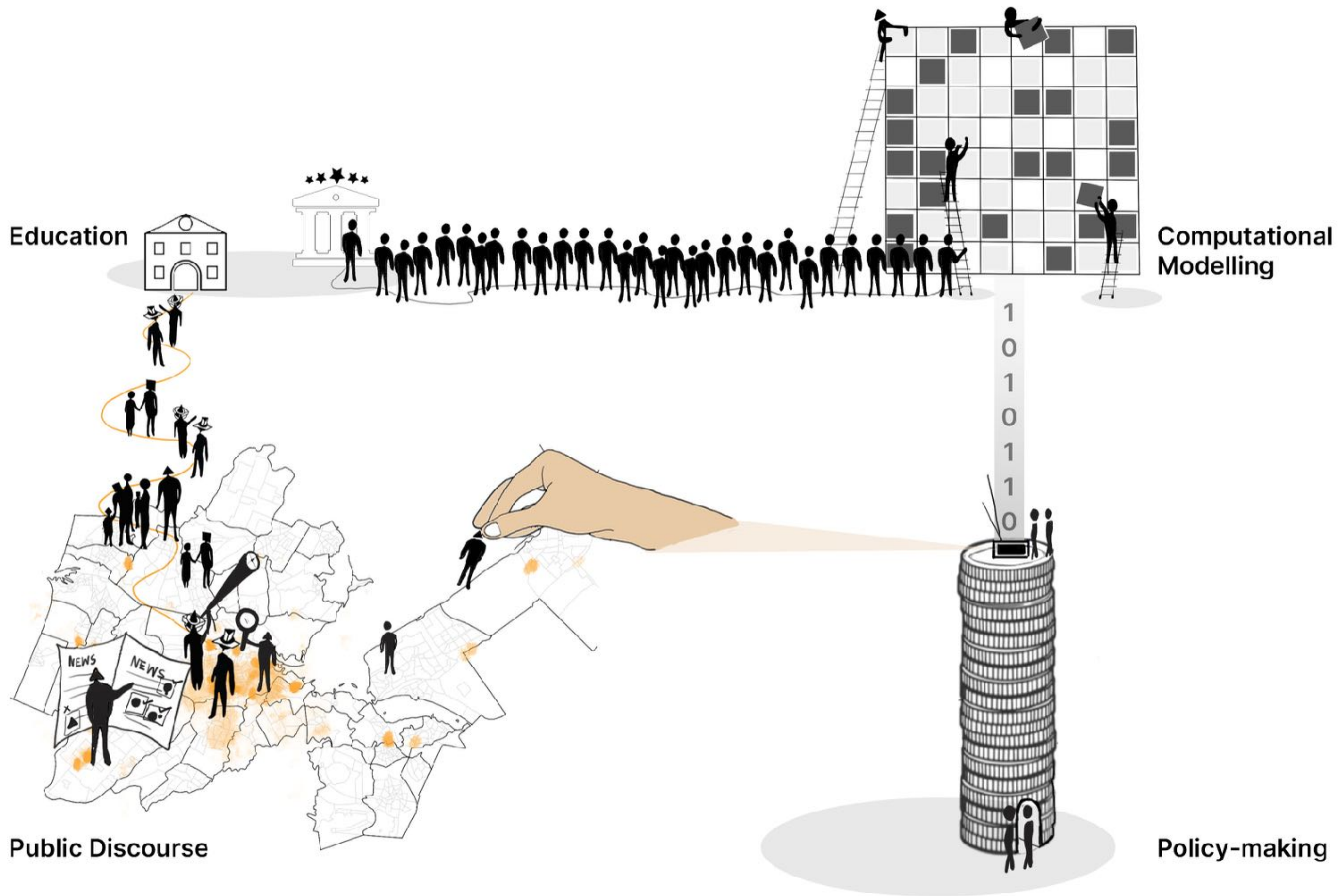
The Schelling model in its current form was published in 1971 by Thomas Schelling in a bid to understand how individual decisions of agents to relocate could lead to a macro pattern of segregation [10]. In the context of widespread racial segregation in the US, the model could demonstrate that for segregation to occur on a macro scale, no deeply entrenched racism was required- even

Urselmans (2018) doi:10.1007/978-3-319-90418-4_1









Our models are reifications of individualistic bias.

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A recurring issue in inequality discourse is the misalignment between philosophical positions of the stated future goal of eradicating inequalities and the proposed pathways to achieve it, often rooted in individualistic meritocracy or individual welfare.

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This is the central predicament of the economy-engineering complex (EEC). **Not all people genuinely believe that all inequalities can - or should - be eliminated.** The deeper issue is not the validity of this philosophical stance (independently of whether or not we agree with it), but its implicit and unconscious character, as often seen within the EEC.

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As Victoria Reyes points out, this predicament can be better understood when considering that the academy is overwhelmingly white, wealthy, male, and Western; as a result, the institution globally lacks the collective experience of oppression. This lack of experience feeds into the misalignment described above, producing an ideological lock-in.

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People who work towards inequality reduction, while ignorant of their own disbelief in the feasibility of this goal, heavily constrain the solution space in which the economy-engineering complex operates.

Nina Simone - Mississippi Goddamn



<https://www.youtube.com/watch?v=LJ25-U3jNWM>

You don't have to live next to me

Nina Simone - Mississippi Goddamn



<https://www.youtube.com/watch?v=LJ25-U3jNWM>

**You don't have to live next to me
Just give me my equality**

Act III: How policy-makers mobilize data and models to reify white supremacy: The case of the Danish 'Parallel Societies Index'

Are You Western?

“Non-western foreign background”, as defined by Statistics Netherlands:

People from Turkey or countries in Africa, Latin America and Asia, with the exception of Former Dutch East Indies/Indonesia and Japan. The first generation consists of people born outside the Netherlands. The second generation are people born in the Netherlands of whom at least one parent was born abroad.

Due to their socioeconomic and cultural position, people from Indonesia and Japan residing in the Netherlands are considered as having a 'western' migration background. These are mainly people born in the former Dutch East Indies and expatriates employed by Japanese companies with their families.

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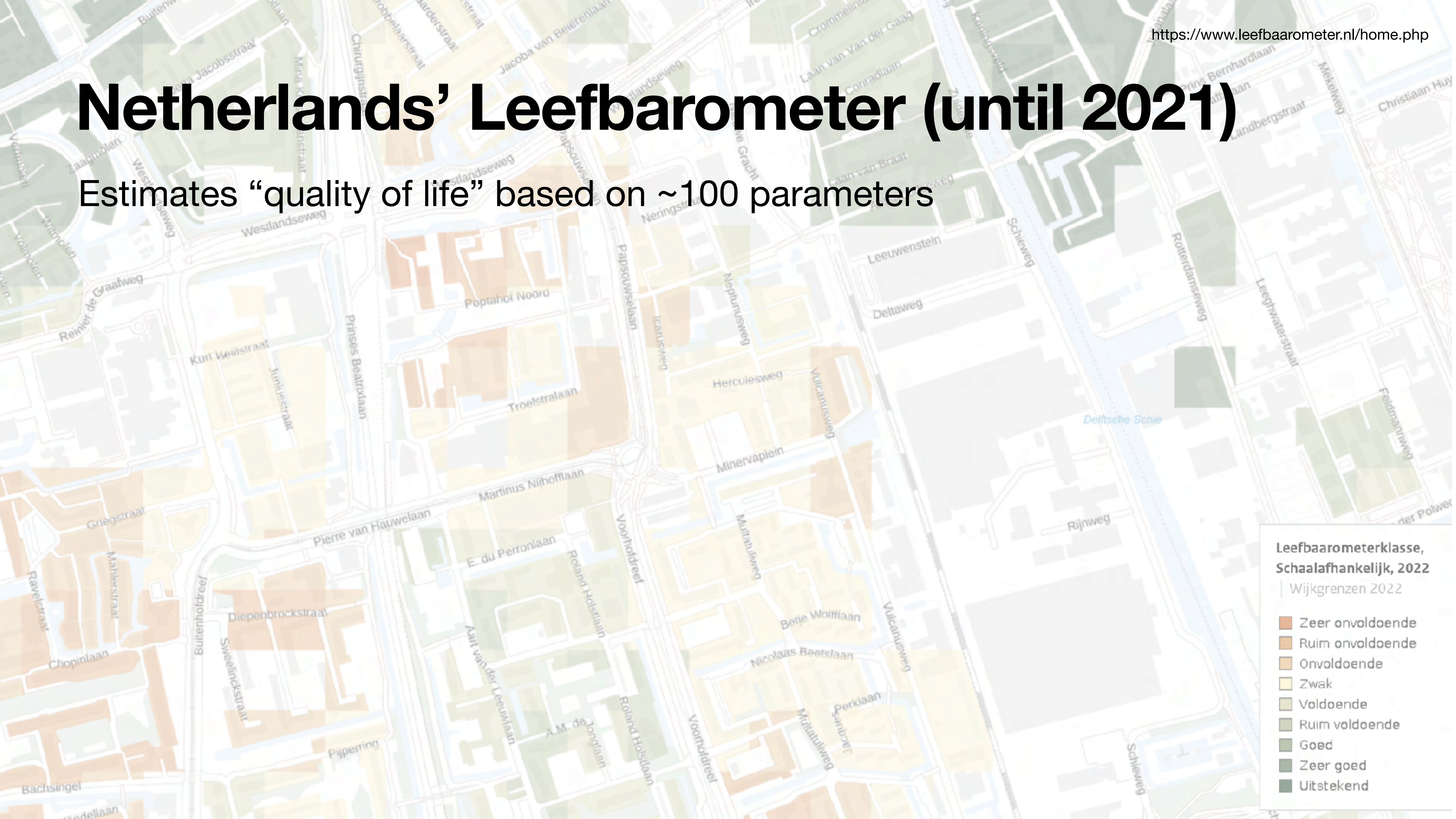
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Netherlands' Leefbaarometer (until 2021)

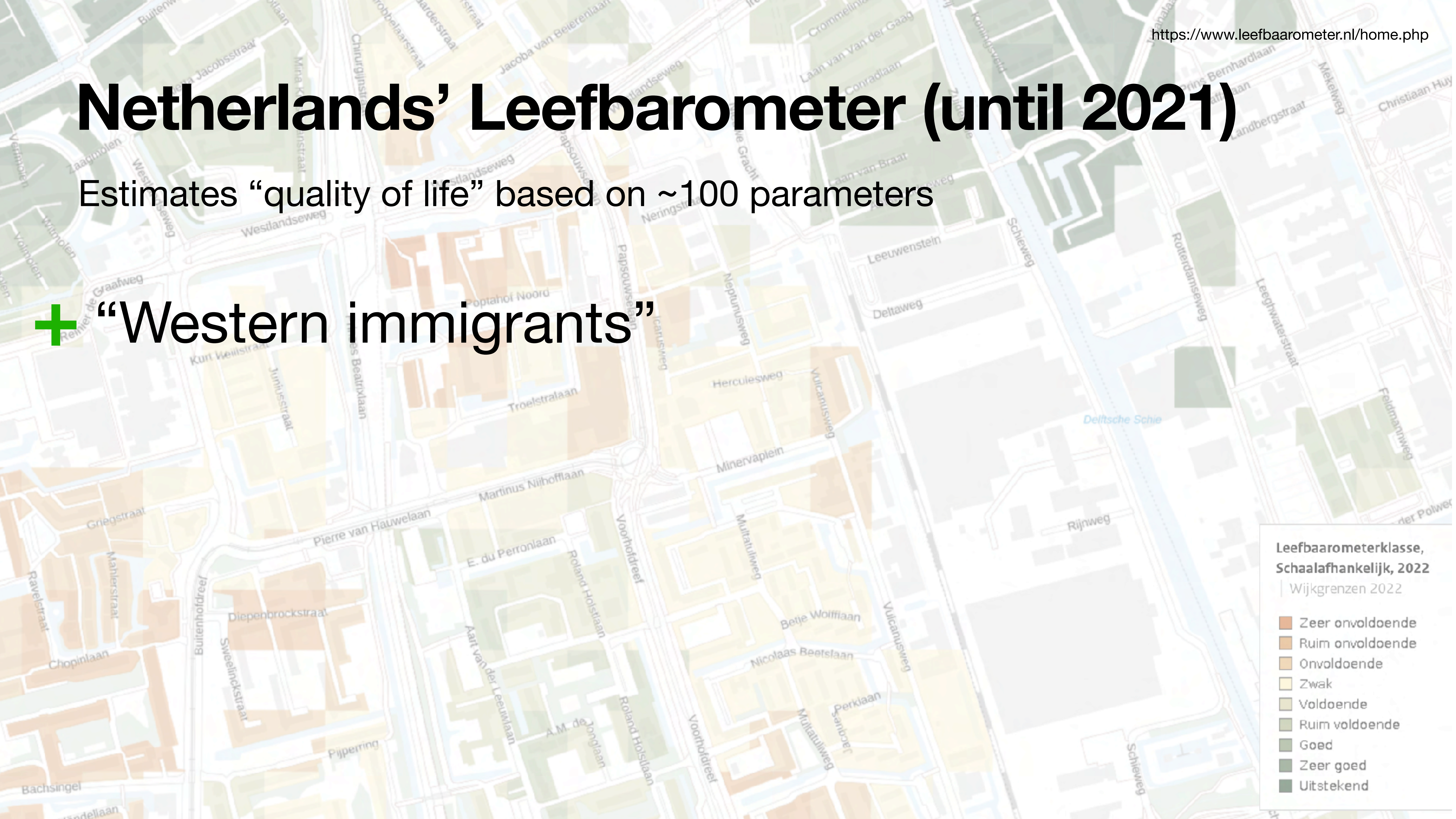
Estimates “quality of life” based on ~100 parameters



Netherlands' Leefbaarometer (until 2021)

Estimates “quality of life” based on ~100 parameters

+ “Western immigrants”



Netherlands' Leefbaarometer (until 2021)

Estimates “quality of life” based on ~100 parameters

+ “Western immigrants”

- “Non-Western immigrants”

- People who receive social benefits

- People who are looking for a job

Leefbaarometerklasse,
Schaalafhankelijk, 2022
| Wijkgrenzen 2022

■	Zeer onvoldoende
■	Ruim onvoldoende
■	Onvoldoende
■	Zwak
■	Voldoende
■	Ruim voldoende
■	Goed
■	Zeer goed
■	Uitstekend

Denmark's 'Ghetto laws'

(rebranded as "Udsatte boligområder og parallelsamfund" in 2021)



Denmark's 'Ghetto laws'

(rebranded as “Udsatte boligområder og parallelsamfund” in 2021)

Percentage of people from
“Non-Western” countries,
& their descendants, and:

- Unemployment rates
- Crime rates
- Education level
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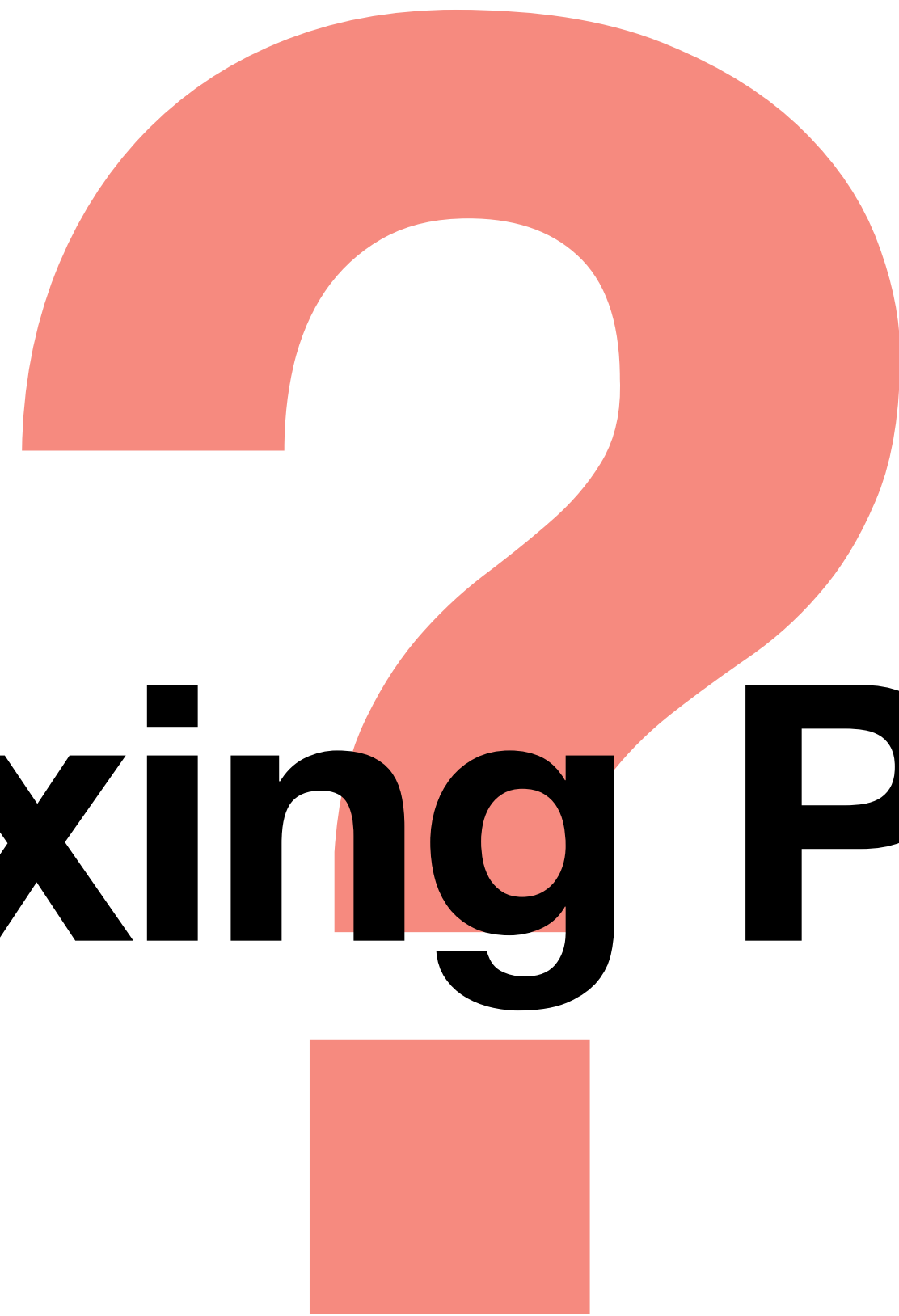
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**Publishing a yearly
updated “Ghetto List”**

**“The first stage of the
development project
is the demolition of
188 homes.”**

Social Mixing Paradigm



8 criteria for “parallel societies”

8 criteria for “parallel societies”

1. Living in a public housing area where there are >25% NWIAD
2. Children aged 1-4 that are not in daycare
3. Children aged 0-6 that are in daycare where >25% NWIAD
4. Children aged 6-16 that are in schools where >25% NWIAD
5. Children aged 12-14 with criminal convictions
6. Children and young adults aged 15-29 with criminal convictions
7. Children and young adults aged 16-29 on welfare from July to December 2014
8. Adults aged 22-59 on welfare for 4 out of 5 years

Act III: How policy-makers mobilize data and models to reify white supremacy: The case of the Danish 'Parallel Societies Index'

anvy@itu.dk

Computational Social Science Against Urban Inequalities A Drama In Three Acts

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